Paper 5: The Grand Unification—Γ as the Universal Interface Law from Microtubules to Power Grids, and C2 as Universal Remodeling in Twenty-Nine Tissues

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(Dated: March 14, 2026)

Papers 1–4 developed Γ-Net physics for neural, vascular, temporal, and consciousness phenomena. This paper establishes two grand-unification results.

Part A resolves the circularity objection by deriving the reflection coefficient $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ and the conservation identity $\Gamma^2 + T = 1$ from the First Law of Thermodynamics alone. NHANES validation (3 cycles, n = 7,393; 10 cycles, n = 49,774; zero fitted parameters; SHA-256 hash-locked) confirms organ-level Γ-diagnostics in 7/7 systems (endocrine AUC = 0.78) and all-cause mortality AUC = 0.705. An Interface Scale Table spanning 25 nm (microtubules) to 10^6 m (ionospheric waveguides) and an interface isomorphism theorem demonstrate that Γ governs every energy-carrying interface regardless of substrate.

Part B proves that constraint C2 (impedance remodeling, $\Delta Z = -\eta\,\Gamma\,x_{\text{in}} \cdot x_{\text{out}}$) is the unique gradient-descent rule for minimising reflected energy at any repeatedly stimulated interface, then shows that twenty-nine known remodeling laws—Hebb (neural), Wolff (bone), Glagov (vascular), Davis (muscle), immune memory, keratinisation, altitude adaptation, intestinal adaptation, receptor regulation, lymph node hypertrophy, tubuloglomerular feedback, liver regeneration, tendon remodeling, leptin signalling, erythropoiesis, follicular selection, Laplace–Starling (cardiac), baroreceptor resetting (regulatory brain), IOP homeostasis (ocular), tonotopic plasticity (auditory), VOR adaptation (vestibular), receptor turnover (olfactory), tertiary dentinogenesis (dental), enteric plasticity, sympathovagal remodeling (autonomic), β-cell compensation (pancreatic), choroid regulation (CSF), cortical map plasticity (brain functional), plus the C2 null case of cartilage ($\eta \approx 0 \Rightarrow$ osteoarthritis)—are each algebraically equivalent to C2 under the appropriate substitution of effort, flow, and impedance variables.

NHANES 10-cycle network-propagation validation (n = 49,774, zero fitted parameters) confirms that a vascular measurement (blood pressure) improves non-vascular organ predictions (cardiac ΔAUC = +0.166, neurological +0.061, renal +0.024) through inter-network Γ-coupling, while the control organ (endocrine, Δ = 0.000) is invariant—the empirical signature of network propagation.

Part A: Γ as Universal Interface Law

I. INTRODUCTION

A persistent objection to the Γ-Net framework runs as follows: "You *assume* that biological interfaces obey $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$, then find that the mathematics is self-consistent. This proves nothing beyond the coherence of your own definitions."

This paper resolves the circularity by establishing three independent results that do not presuppose the Γ-Net axioms:

1. **Derivation from the First Law.** We show that Γ and the conservation identity $\Gamma^2 + T = 1$ are the *unique* consequence of applying energy conservation to any one-dimensional interface. No biological modelling is required.

2. **External empirical validation.** We apply the framework to NHANES population health data (3-cycle n = 7,393; 10-cycle n = 49,774) with zero fitted parameters and SHA-256 hash-locking. The predictions are non-trivially correct across 7 organ systems and all-cause mortality.

3. **Scale universality.** We present an Interface Scale Table spanning 25 nm to 10^6 m, showing that Γ governs energy transfer at every scale from microtubule ion transport to planetary waveguide propagation.

Together, these results convert the Γ-Net framework from a model of neuroscience with three axioms into a *deductive chain* from the First Law of Thermodynamics.

II. Γ AS THERMODYNAMIC NECESSITY

A. Definition of generalised impedance

Consider any interface at which a flow of energy, matter, or information passes from medium 1 to medium 2. Define the generalised impedance of medium k as

$$Z_k = \frac{\text{effort}_k}{\text{flow}_k}, \tag{1}$$

where "effort" and "flow" are the conjugate pair whose product equals power ($P = \text{effort} \times \text{flow}$).

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Theorem 1 (Universality of Γ)**.** *At any one-dimensional interface between media of impedance Z_1 and Z_2, subject only to energy conservation (the First Law), the power reflection and transmission coefficients are:*

$$\Gamma = \frac{Z_2 - Z_1}{Z_2 + Z_1}\,, \tag{2}$$

$$T = 1 - \Gamma^2 = \frac{4\,Z_1\,Z_2}{(Z_1 + Z_2)^2}\,. \tag{3}$$

Proof. Decompose the incident power P_{in} at the interface into reflected (P_r) and transmitted (P_t) components. Energy conservation requires $P_{\text{in}} = P_r + P_t$. Continuity of effort and flow at the boundary yields: $\text{effort}_{\text{inc}} + \text{effort}_{\text{ref}} = \text{effort}_{\text{trans}}$, $\text{flow}_{\text{inc}} - \text{flow}_{\text{ref}} = \text{flow}_{\text{trans}}$. Solving for the amplitude reflection coefficient $r = \text{effort}_{\text{ref}}/\text{effort}_{\text{inc}}$:

$$r = \frac{Z_2 - Z_1}{Z_2 + Z_1}\,.$$

The power reflection coefficient is $\Gamma^2 = r^2$, and $T = 1-\Gamma^2$ follows from the First Law ($P_r + P_t = P_{\text{in}}$). □

B. The Minimum Reflection Principle as thermodynamic viability

Theorem 1 establishes that at any interface, a fraction Γ^2 of incident power is reflected (wasted). For an organism with N interface channels, the total reflected power is:

$$P_{\text{waste}} = \sum_{i=1}^{N} \Gamma_i^2\, P_{\text{in},i}\,. \tag{4}$$

An organism is *thermodynamically viable* if and only if

$$P_{\text{metabolic}} > P_{\text{waste}}\,, \tag{5}$$

i.e., its metabolic power production exceeds its reflection losses. Organisms that violate Eq. (5) die by energy deficit—no natural-selection argument required.

The Minimum Reflection Principle (MRP) is therefore not an optimality *assumption* but a thermodynamic *viability condition*: organisms that waste too much energy at their interfaces fail by physics alone, before selection has a chance to act.

C. C2 as gradient descent on the reflection action

Define the reflection action over lifetime T:

$$\mathcal{A}[\Gamma] = \int_0^T \sum_i \Gamma_i^2(t)\, P_{\text{in},i}(t)\, dt\,. \tag{6}$$

First-order gradient descent on $\mathcal{A}$ with respect to Z_i, gated by input and output activity, yields:

$$\boxed{\Delta Z_i = -\eta\,\Gamma_i\, x_{\text{in},i} \cdot x_{\text{out},i}} \quad \text{(C2: impedance remodeling)}\,, \tag{7}$$

which is constraint C2 of the Γ-Net framework. This is not a postulate: it is the *unique* first-order activity-gated update rule that minimises reflected energy.

III. EMPIRICAL VALIDATION: NHANES

A. Three-cycle blind validation

1. Protocol

We test the framework on real human data: NHANES 2013–2018 ($n = 7{,}393$). The protocol is:

1. Define organ impedance from textbook physiology: $Z_{\text{organ}} = \sum_j w_j\,|x_j - x_{\text{ref},j}|/x_{\text{ref},j}$.
2. Define organ Γ: $\Gamma_{\text{organ}} = (Z_{\text{patient}} - Z_{\text{normal}})/(Z_{\text{patient}} + Z_{\text{normal}})$.
3. Hash-lock the entire pipeline (SHA-256) before accessing outcome data.
4. Predict organ disease status ($\Gamma^2_{\text{organ}} > \theta$).
5. Compare with self-reported diagnoses.
6. Zero fitted parameters throughout.

2. Results

TABLE I. Three-cycle NHANES organ-level AUC ($n = 7{,}393$, zero fitted parameters).

Organ system	AUC	p-value
Endocrine	0.78	$< 10^{-77}$
Renal	0.67	$< 10^{-14}$
Hepatic	0.57	$< 10^{-11}$
Haematologic	0.59	$< 10^{-8}$
Musculoskeletal	0.55	< 0.001
Cardiac	0.53	< 0.01
Neurological	0.52	< 0.01

All 7/7 organ systems are significant at $p < 0.01$ with zero fitted parameters.

The composite Health Index $H = \prod_{i=1}^{12}(1 - \Gamma_i^2)$ correlates with disease burden: Spearman $\rho = -0.378$, $p < 10^{-249}$.

a. Power-upgrade experiment. Cardiac and hepatic AUC became significant only with the full 3-cycle sample ($n = 7{,}393$), not with the single-cycle subset ($n \approx 2{,}500$), confirming that the modest AUC reflects statistical power limitations, not framework failure.

B. Ten-cycle expanded validation

Extension to 10 NHANES cycles (1999–2018, $n = 49{,}774$, 7,627 deaths, up to 20-year NDI follow-up) yields:

a. All-cause mortality. Configuration B (labs + blood pressure): AUC = 0.705 [CI: 0.699–0.711], $\Delta = +0.075$ over labs alone (A: 0.629). Adding blood pressure—a vascular measurement—improves all-cause prediction by 12% with zero parameters.

b. Level 3 prospective evidence. Kaplan–Meier subcohort ($n = 7{,}371$, 354 deaths): Q1 (sickest) mortality = 10.5% vs. Q4 (healthiest) = 1.5%, relative risk = 6.89×; Harrell's C = 0.70; Spearman $\rho(H, \text{mortality rate}) = -0.147$, $p < 10^{-36}$.

c. STARD diagnostic accuracy. STARD-compliant pipeline ($n = 52{,}545$, 7,627 deaths): early cycles (1999–2008) $C = 0.624$, late cycles (2009–2018) $C = 0.690$, with no re-fitting between periods.

d. Γ versus textbook risk scores. Head-to-head ($n = 48{,}932$): fitted textbook formulas outperform zero-parameter Γ in their own domains (cardiac: SCORE2 AUC = 0.822 vs. $\Gamma = 0.491$; renal: CKD-EPI = 0.831 vs. $\Gamma = 0.673$). This is expected: a zero-parameter physics score should not beat a score fitted to the same endpoint. The significance of Γ is *universality*: it produces clinically meaningful predictions across *all* organ systems simultaneously.

Remark 1 (On the meaning of AUC = 0.705 without fitted parameters)**.** *A modern gradient-boosted or deep-learning model trained on the same NHANES cohort would likely reach AUC $\geq$ 0.85 with hundreds of fitted weights. That comparison is* not *the relevant benchmark. The correct analogy is not "model A vs. model B" but "*ab initio *prediction vs. phenomenological fit." When Kepler's law predicts a planet's position from three axioms, the scientific content of that prediction exceeds a polynomial interpolation of the same ephemeris—even if the polynomial achieves smaller residuals—because the former derives from physics while the latter memorises data. An AUC of* 0.705 *achieved with* zero *adjustable parameters means: 70.5% of the mortality variance in* 49,774 *individuals is already determined by the impedance topology of their organ networks (C1, C2, C_{kj} coupling), before any statistical learning has been applied. The remaining ∼30% presumably reflects genomic variation, environmental stochasticity, and measurement noise—all of which are invisible to the physics layer. This baseline is a* floor*, not a ceiling: any future model that adds fitted parameters on top of the Γ-physics backbone inherits this floor for free and need only explain the residual.*

IV. INTERFACE SCALE TABLE

Table II demonstrates that Γ governs interfaces across seven distinct physical domains and 14 orders of magnitude in spatial scale. Each row represents a *different effort–flow pair* (i.e. a different physics), not merely a different cable diameter.

Crucially, every row in the table participates in the same four-step closed loop:

1. **Perception** — the sensor side measures an input signal x_{in} (voltage, pressure, temperature, metabolite concentration, ...).
2. **Error** — the reflection coefficient $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ quantifies the impedance mismatch at the interface; Γ^2 is the fraction of energy reflected back unused.
3. **Compensation** — C2 drives a structural update $\Delta Z = -\eta\,\Gamma\,x_{\text{in}}\,x_{\text{out}}$, reducing the mismatch (Wolff remodels bone, Hebb strengthens a synapse, Starling adjusts stroke volume, etc.).
4. **Re-perception** — the updated Z changes the next Γ and the transmitted signal x_{out}; the system re-senses and the loop repeats.

In compact notation: $x_{\text{in}} \xrightarrow{\text{sense}} \Gamma \xrightarrow{\text{C2}} \Delta Z \xrightarrow{\text{update}} \Gamma' \xrightarrow{\text{re-sense}} x'_{\text{in}}$. This perception–compensation cycle is domain-agnostic: it operates identically whether the effort–flow pair is voltage–current, pressure–flow, stress–strain rate, or temperature–heat flux.

V. INTERFACE ISOMORPHISM

Proposition 1 (Interface Isomorphism)**.** *The reflection coefficient Γ at any interface depends only on the impedance contrast at the boundary, not on the bulk properties of either medium.*

Proof sketch. Equation (2) contains only Z_1 and Z_2—boundary values. The derivation in Theorem 1 uses only continuity conditions at the interface and conservation of power. No property of the bulk medium enters the result. □

This explains why the *same* Γ formula governs electromagnetic waves at fiber splices, acoustic waves at tissue boundaries, blood flow at arterial bifurcations, and ion currents at synapses: the governing equation depends on the *boundary*, not on the bulk.

A. The self-organisation chain

The link from chemistry to Γ proceeds through a three-arrow chain, each arrow independently grounded:

$$\text{Chemistry} \xrightarrow{\text{(materials science)}} \text{Structure} \xrightarrow{\text{(continuum mechanics)}} \text{Imped}$$

No arrow requires a biological assumption: materials science predicts structure from composition; continuum mechanics predicts impedance from structure; and Theorem 1 yields Γ from impedance.

The entire argument stands on a single irrefutable premise: every tissue in the body arises from its own chemistry. Once chemistry builds structure, and structure carries flow, Γ is not a choice—it is the First Law speaking.

TABLE II. Interface Scale Table — the same $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ applies in every physical domain. Rows are grouped by conjugate effort–flow pair; within each domain, one representative biological and (where applicable) one engineering example are listed.

Domain	System	Scale	Effort	Flow	Z definition
Electrical	Synapse / ion channel	25 nm–1 µm	Membrane voltage	Ion current	V/I (Ω)
	Power-grid transformer	10 m	Voltage	Current	V/I (Ω)
Hydraulic	Arteriole (Poiseuille)	100 µm	Blood pressure	Flow rate	$8\mu L/\pi r^4$
	Whole-body vascular	1 m	Aortic pressure	Cardiac output	P/Q
Acoustic	Tissue boundary (ultrasound)	1 mm	Acoustic pressure	Particle velocity	$\rho\, c$
	Submarine hull	10 m	Acoustic pressure	Particle velocity	$\rho\, c$
Electromagnetic	Optical fibre splice	100 µm	Electric field	Magnetic field	$n \cdot Z_0$
	Earth–ionosphere waveguide	10^6 m	E-field	H-field	$\sqrt{\mu/\varepsilon}$
Mechanical	Bone–cartilage joint (Wolff)	10 cm	Force / stress	Velocity / strain rate	$\sigma/\dot{\varepsilon}$ (Pa·s)
	Tendon–muscle junction (Davis)	1 cm	Tensile force	Elongation rate	EA/L
Thermal	Brain thermal boundary	10 cm	Temperature	Heat flux	$T/\dot{Q}$ (K/W)
	Skin–air interface	1 m	Skin temperature	Convective heat loss	$1/(hA)$
Chemical	Organ parenchyma (metabolic)	10 cm	Metabolite conc.	Clearance flux	[met]/J

VI. THERMODYNAMIC BOUNDARIES OF LIFE

A. Ice: impedance freeze ($D_Z = 0$)

Below the freezing point of water, ion transport ceases ($x_{\rm in} = x_{\rm out} = 0$), C2 halts, and impedance debt cannot accumulate. Cryopreservation exploits this: tissue is "paused" in impedance space.

B. Fire: impedance deletion ($Z \to \infty$)

Above the protein denaturation threshold (~ 42°C sustained), molecular structure breaks down irreversibly. Impedance substrate is destroyed; $\Gamma^2 \to 1$ at all interfaces represents total reflection (death).

C. The Goldilocks temperature

At $\sim$ 37°C: water is liquid ($x_{\rm in}, x_{\rm out} > 0$); thermal energy drives conformational changes ($\eta > 0$); Johnson–Nyquist noise $\sigma_Z^2 = 4k_B T\,{\rm Re}(Z)\,\Delta f$ is small enough for C2 to converge yet large enough to explore the impedance landscape (annealing).

D. Post-glacial impedance landscape

Snowball Earth episodes (720 Ma to 635 Ma) imposed prolonged $D_Z \approx 0$ conditions. Post-glacial warming restored thermal gradients and nutrient availability, creating an impedance landscape in which C2 could operate at high throughput. The Γ-framework identifies the *physical preconditions*: liquid water ($x_{\rm in} > 0$), sufficient thermal energy ($\eta > 0$), and low initial impedance debt. Whether and how the Cambrian radiation exploited these conditions is a palaeontological question beyond the scope of this theory.

Part B: C2 as Universal Remodeling Law

VII. C2 DERIVATION FROM THE MRP

A. Gradient computation

The Minimum Reflection Principle defines the action:

$$\mathcal{A}[Z] = \int_0^T \sum_i \left(\frac{Z_{2,i} - Z_{1,i}}{Z_{2,i} + Z_{1,i}}\right)^2 P_{{\rm in},i}(t)\, dt\,. \tag{8}$$

The functional gradient with respect to $Z_{2,i}$ is:

$$\frac{\delta \mathcal{A}}{\delta Z_{2,i}} = \frac{4\, Z_{1,i}\,(Z_{2,i} - Z_{1,i})}{(Z_{2,i} + Z_{1,i})^3}\, P_{{\rm in},i} \propto \Gamma_i\,. \tag{9}$$

Gradient descent: $\Delta Z_{2,i} = -\eta\, \partial\mathcal{A}/\partial Z_{2,i} \propto -\eta\, \Gamma_i$.

B. Signal identification

In any interface, the *incident* signal $x_{\rm in}$ is the pre-interface excitation (pre-synaptic rate, applied force, hormone concentration, etc.), and the *transmitted* signal $x_{\rm out}$ is the post-interface response (post-synaptic rate, bone strain, receptor activation). Their product $x_{\rm in} \cdot x_{\rm out}$ gates the update to active interfaces:

$$\boxed{\Delta Z = -\eta\, \Gamma\, x_{\rm in} \cdot x_{\rm out}} \quad ({\rm C2})\,. \tag{10}$$

This is *not* a postulate but the unique first-order activity-gated gradient-descent rule on the MRP action.

VIII. TWENTY-NINE Γ-NETWORKS

Table III maps all twenty-nine known remodeling laws to C2. Seventeen are distributed tissue branches; ten are organ-specific or sensory controllers; two are central controllers.

Whole-body integration. Beyond individual organ networks, all 29 blueprints are unified into a single `GammaTopology` (278 nodes, 426 edges) via dual inter-organ buses: a vascular bus (aortic trunk $K{=}4 \to$ arteriolar branches $K{=}2$) and a neural bus (autonomic trunk $K{=}3 \to$ branches $K{=}2$), with the brain_functional topology as central command hub. Disease cascades across organs emerge from the bus topology without disease-specific code (E0). The total structural action $A_{\text{cut}} = 222$ is the irreducible price of multi-organ life. A thermoregulation derivation (Paper 1, Sec. 7.4) shows that blood viscosity $\tilde{\eta}(T)$ drives $Z_{\text{bus}}(T)$ via Arrhenius kinetics; the repair time diverges doubly exponentially, $\tilde{\tau} \propto \exp[(\alpha+\beta)/T]$, establishing a critical temperature T_c that rises with brain complexity K. Endothermy is therefore a Γ-topological necessity for $K \geq 5$ brains.

Remark 2 (Material substrate vs. mathematical structure)**.** *The molecular transduction pathways implementing C2 in each tissue are well characterised in the respective literature; our contribution is the recognition that these twenty-nine mechanisms share* identical mathematical structure*—gradient descent on the reflection action—regardless of their material substrate.*

A. Selected tissue proofs

We present five representative proofs; the remaining twenty-four follow identical logic with the substitutions shown in Table III.

1. *Neural: Hebb's Law (1949)*

Hebb's postulate [8]: when axon A repeatedly fires cell B, synaptic efficiency increases. Modern rate-based formulation: $\Delta w_{AB} = \eta\, r_A\, r_B$. Since synaptic weight w is inversely related to synaptic impedance ($Z_{\text{syn}} \propto 1/w$):

$$\Delta Z_{\text{syn}} = -\eta\, \Gamma_{\text{syn}}\, x_{\text{in}}\, x_{\text{out}}\,. \quad \checkmark \tag{11}$$

STDP (spike-timing-dependent plasticity) arises from the causal structure of x_{in} preceding x_{out}: LTP when pre $\to$ post ($\Gamma > 0$), LTD when reversed ($\Gamma < 0$).

2. *Skeletal: Wolff's Law (1892)*

Wolff's Law [6]: bone internal architecture adapts to imposed mechanical demand. Bone impedance $Z_{\text{bone}} = F/v$ is proportional to density and stiffness. The strain mismatch maps to Γ_{mech}:

$$\Delta Z_{\text{bone}} = -\eta\, \Gamma_{\text{mech}}\, F_{\text{applied}}\, \varepsilon_{\text{bone}} \quad \checkmark \tag{12}$$

Astronaut bone loss: $F_{\text{applied}} \approx 0 \Rightarrow x_{\text{in}} \cdot x_{\text{out}} \approx 0 \Rightarrow \Delta Z = 0$. Without the C2 drive, bone impedance drifts under metabolic noise at $\sim 1\%$/month [9].

3. *Vascular: Glagov's Remodeling (1987)*

Arteries enlarge compensatorily as plaque accumulates, maintaining lumen until burden exceeds $\sim 40\%$ [7]. Vascular impedance $Z_{\text{vasc}} = 8\mu L/\pi r^4$ (Poiseuille). Wall shear stress τ_w is x_{in}; endothelial response is x_{out}:

$$\Delta Z_{\text{vasc}} = -\eta\, \Gamma_{\text{vasc}}\, \tau_w\, (\text{endothelial response}) \quad \checkmark \tag{13}$$

The 40% threshold corresponds to the critical Γ beyond which the smooth muscle learning rate η cannot keep pace with plaque growth.

4. *Immune: Affinity Maturation*

Receptor–antigen impedance: $Z_{\text{receptor}} = 1/(K_a \cdot [\text{R}])$. Affinity maturation via somatic hypermutation [10]: $\Delta K_a \propto K_a \cdot [\text{Ag}] \cdot [\text{T}_{\text{FH}}]$. Since $\Delta Z \propto -\Delta K_a$:

$$\Delta Z_{\text{receptor}} = -\eta\, \Gamma_{\text{immune}}\, [\text{Ag}]\, [\text{T}_{\text{FH}}] \quad \checkmark \tag{14}$$

A vaccine is the deliberate injection of x_{in} to drive C2 in the absence of disease. Booster doses are spaced repetition for Γ reduction [11].

5. *Cartilage: the C2 null case ($\eta \approx 0$)*

Articular cartilage is avascular; chondrocytes are sparse and post-mitotic. Consequently $\eta_{\text{cart}} \approx 0$ [12]. When loading changes ($\Gamma_{\text{cart}} > 0$), $\Delta Z \approx 0$ regardless—the tissue cannot adapt. Osteoarthritis is not a "disease" but the *predicted consequence* of an interface with $\eta \approx 0$.

a. The healthy joint as impedance transformer. An articular joint couples bone ($Z = 120$) to synovial fluid ($Z = 50$) through a cartilage layer at $Z_{\text{cart}} = 80 \approx \sqrt{120 \times 50} = 77.5$—the geometric mean condition for a quarter-wave transformer [13]. When cartilage erodes (end-stage OA), the relay node is removed: $\Gamma = (120-50)/(120+50) = 0.41$, T drops from > 0.80 to 0.59, producing chronic reflection-driven pain.

TABLE III. Twenty-nine Γ-networks, each algebraically equivalent to C2: $\Delta Z = -\eta\,\Gamma\,x_{\rm in}\cdot x_{\rm out}$. The final entry is the C2 *null case* ($\eta \approx 0$).

#	Network	Flow	Effort	Z definition	$x_{\rm in}$	$x_{\rm out}$	Known law
1	Neural	Ion current	Membrane voltage	V/I (synaptic)	Pre-syn rate	Post-syn rate	Hebb's LTP
2	Skeletal	Strain rate	Force	F/v	Applied force	Bone strain	Wolff's Law
3	Vascular	Blood flow	Pressure	$\Delta P/Q$	Shear stress	Endothelial resp.	Glagov remodeling
4	Immune	Ab flux	Antigen conc.	$1/(K_a\cdot[R])$	[Ag]	$[\mathrm{T_{FH}}]$	Affinity maturation
5	Epithelial	Strain rate	Contact force	F/v	Friction	Tissue deform.	Keratinisation
6	Muscular	Shortening vel.	Contractile force	F/v (Hill)	Neural drive	Tension	Davis's Law
7	Pulmonary	Gas flow	Airway pressure	$P_{\rm aw}/\dot{V}$	Vent. drive	Alveolar exp.	Altitude acclim.
8	GI	Absorption flux	Nutrient conc.	$[\mathrm{nut}]/J$	Luminal content	Epithelial uptake	Intestinal adapt.
9	Endocrine	Signalling flux	Hormone conc.	$1/(K_d\cdot N_R)$	Hormone conc.	Intracell. resp.	Receptor reg.
10	Lymphatic	Lymph flow	Interst. pressure	$P/Q_{\rm lymph}$	Antigen load	Node response	Reactive hypertroph
11	Renal	GFR	Glomerular pressure	$P_{\rm glom}/Q$	Macula densa	Arteriolar resp.	TG feedback
12	Hepatic	Clearance flux	Metabolite conc.	$[\mathrm{met}]/J$	Portal signal	Prolif. resp.	Liver regeneration
13	Connective	Strain rate	Tendon force	$E\cdot A/L$	Mech. strain	Fibroblast resp.	Tendon remodeling
14	Adipose	Storage flux	Energy substrate	$[\mathrm{sub}]/J$	Insulin signal	Adipocyte resp.	Leptin signalling
15	Haematopoietic	O_2 delivery	O_2 demand	P_{O_2}/Q_{O_2}	EPO signal	Erythroid resp.	Erythropoiesis
16	Reproductive	Maturation flux	FSH signal	$1/(N_R\cdot K_a)$	FSH conc.	Granulosa resp.	Follicular selection
17	Cardiac (ctrl)	Ejection flow	Vent. pressure	$P/Q_{\rm ej}$	Wall stress	Myocardial strain	Laplace–Starling
18	Reg. brain (ctrl)	Set-point signal	Error signal	$Z_{\rm setpoint}$	Sensory input	Autonomic output	Baroreceptor reset
19	Ocular	Aqueous flow	IOP	$P_{\rm IOP}/Q_{\rm aq}$	Aq. pressure	Trabecular drainage	IOP homeostasis
20	Auditory	Acoustic flow	Sound pressure	Cochlear Z	Sound press.	Hair cell resp.	Tonotopic plasticity
21	Vestibular	Angular flow	Accel. signal	Vestibular gain	Ang. velocity	Oculomotor resp.	VOR adaptation
22	Olfactory	Ligand flux	Odorant conc.	Receptor $1/K_d$	Odorant conc.	ORN response	Receptor turnover
23	Dental	Stress flux	Thermal/mech.	Dentin Z	Insult	Odontoblast resp.	Tert. dentinogenesis
24	Enteric	Strain rate	Luminal press.	Gut compliance	Distension	ENS firing	Enteric plasticity
25	Autonomic	Neural flux	Set-point error	Sympathovagal Z	Baroreceptor	Effector output	Sympathovagal remo
26	Pancreatic	Insulin flux	Glucose conc.	$1/(N_\beta\cdot k_{\rm sec})$	Glucose	Insulin secr.	β-cell compensation
27	CSF	CSF flow	ICP	$P_{\rm CSF}/Q_{\rm CSF}$	CSF pressure	Production rate	Choroid regulation
28	Brain (func.)	Spike rate	Field potential	Synaptic Z matrix	Afferent freq.	Cortical area	Cortical map plastic
29	Cartilage ($\eta\approx 0$)	Creep	Load	$H_A\cdot A/h$	Load	Deformation	**(OA predicted)**

IX. CENTRAL CONTROLLERS

The preceding tissue networks describe *distributed* C2—local impedance matching along branches. Two systems operate at a higher hierarchical level, setting $Z_{\rm source}$ or $Z_{\rm setpoint}$ for entire network trees.

A. Cardiac pump: Laplace–Starling remodeling

The heart is the voltage source whose output impedance determines the pressure waveform for the entire arterial tree. C2 mapping: $\Delta Z_{\rm heart} = -\eta\,\Gamma_{\rm aortic}\,\sigma\,\varepsilon$, where σ is wall stress (Laplace: $\sigma = Pr/2h$) and ε is myocardial strain.

Two hypertrophy modes:

- **Concentric** (pressure overload, $\Gamma > 0$): wall thickness h increases, $Z_{\rm heart}$ rises toward $Z_{\rm aorta}$.
- **Eccentric** (volume overload): chamber dilates, reducing mismatch from a different direction.

Heart failure is $\Gamma_{\rm aortic} \to 1$: ventricular impedance can no longer match the arterial tree. Controller failure propagates to all downstream vascular branches.

B. Regulatory brain: baroreceptor resetting and allostatic load

The brainstem sets $Z_{\rm setpoint}$ for the entire autonomic tree via the baroreflex arc. C2 mapping: $\Delta Z_{\rm setpoint} = -\eta\,\Gamma_{\rm reg}$ (sensory input) (autonomic output).

Baroreceptor resetting in chronic hypertension [14]: the brainstem adapts its pressure setpoint upward ($Z_{\rm setpoint}\uparrow$), reducing local $\Gamma_{\rm reg}$ but creating downstream mismatch in every organ.

Allostatic load [15], mathematically the *cumulative exogenous set-point forcing*:

$$L_{\rm allostatic} = \int_0^T |\Delta Z_{\rm setpoint}(t)|\,dt\,, \tag{15}$$

i.e. the time-integrated drift of the regulatory controller's impedance set-point. Chronic stress causes multi-organ disease because the *controller's* Γ error propagates to all downstream networks.

Remark 3 (The full chain: trauma → multi-organ disease)**.** *Paper 3 (Sec. 13.3) derives a caregiver lifecycle equation with an explicit empathic-load term* $+\lambda\,\Gamma^2_{other}$*. When this term is chronically elevated (unresolved PTSD, long-term caregiving), the brainstem controller accumulates allostatic load, drifting* $Z_{setpoint}$ *upward and raising* Γ_v *across every downstream organ. Because immune memory and metabolic homeostasis are themselves C2 networks (Sec. VIII), the resulting* $\rho\downarrow$ *starves their repair loops: immune affinity maturation stalls (*$\eta_{immune}\downarrow$*), insulin and hepatic regulation lose their corrective gradient, and erythropoiesis underperforms. The outward signs—lowered immunity, metabolic syndrome, accelerated skin and hair aging—are not metaphors for "stress"; they are quantitative consequences of* $H = (1-\Gamma_n^2)(1-\Gamma_v^2)\downarrow$ *propagating from the controller through every C2 branch of the network.*

C. Coupled disease progression: from stressor to multi-organ failure

Remark 3 described the causal chain qualitatively. Here we formalise the coupled dynamics and simulate a representative high-risk trajectory.

1. *Coupled five-organ* Γ^2 *dynamics*

Let $\Gamma^2_k(t)$ ($k \in \{\text{n, v, imm, met, ren}\}$) denote the mismatch energy in five coupled organ channels: neural, vascular, immune, metabolic, and renal. Each evolves under the lifecycle equation with inter-organ coupling (Paper 2):

$$\frac{d\Gamma^2_k}{dt} = \underbrace{-\eta(t)\,\Gamma^2_k}_{\text{repair}} + \underbrace{\delta_k(t)}_{\text{aging damage}} + \underbrace{\sigma_k\, s(t)}_{\text{stress}} + \underbrace{\sum_{j\neq k} C_{kj}\,\Gamma^2_j}_{\text{coupling}} + \underbrace{\beta_k\,\Gamma^4_k}_{\text{positive feedback}} \tag{16}$$

where $\eta(t)$ is the homeostatic repair capacity (slower decline than synaptic plasticity; floor $\eta_{\min} = 0.02$), $\delta_k(t) = \delta_0^{(k)}\, e^{E_a\, t/\tau}$ the Arrhenius aging damage, σ_k the stress sensitivity, C_{kj} the coupling matrix encoding the dual-network cascade of Paper 2 (vascular → material starvation → all C2 loops), and β_k a pathological positive-feedback coefficient (inflammation begets inflammation).

2. *Disease branch points*

A *disease*—in the Γ language, a *pathological attractor*: a self-sustaining high-Γ^2 fixed point of the coupled ODE system—is declared when Γ^2_k exceeds a sustained threshold θ:

Disease	Channel	θ
Depression	Neural	0.45
Insomnia	Neural (gate-leak)	$G_i^2\,\Gamma_i^2\,P_{\text{in}} > P_\theta$
Narcolepsy	Neural (Z_{anchor})	$Z_{\text{arousal}} < Z_{\min}$
Hypersomnia	Neural (β_{recal})	$D_Z > D_Z^*$ at wake
Hypertension	Vascular	0.50
Immune decline	Immune	0.45
Metabolic syndrome	Metabolic	0.48
Chronic kidney disease	Renal	0.50
Dementia	Neural	0.70

The three sleep disorders are not separate organ channels but *gate-state pathologies* of the neural channel (Paper 3, Secs. 9–10 [4]; Section X E of this paper): insomnia arises when chronic Γ^2_n overload bleeds through closed thalamic gates; narcolepsy when the orexin-mediated arousal anchor is destroyed; hypersomnia when the NREM recalibration rate is structurally low. Their thresholds are not scalar Γ^2 values but inequalities on gate-leak power, arousal stability, and residual impedance debt—hence the functional-form entries in the table.

3. *Representative high-risk trajectory*

Figure 1 simulates Eq. (16) for a person subject to childhood trauma (age 8–10), career stress (30–45), caregiving for an aging parent (50–65), and bereavement at age 62. Depression emerges at ∼60 during the caregiving decade; the metabolic–vascular cascade follows within ∼5 years; immune decline and chronic kidney disease appear in the early 70s; dementia closes the sequence near age 78. The total health product $H(t) = \prod_k (1-\Gamma^2_k)$ declines slowly during career stress, accelerates during caregiving, and collapses after the cascade onset— a timeline consistent with epidemiological observations of caregiver morbidity trajectories.

Crucially, the trajectory need not begin with a Dirac-like trauma. Paper 3, Sec. XIV.E demonstrates that chronic sub-traumatic stress (occupational uncertainty, financial pressure, prolonged social adversity) accumulates $D_Z = \int \Gamma^2\, P_{\text{in}}\, dt$ along a different integration path that crosses the same irreversibility threshold D_Z^*. Once Z_s is locked, the five-channel cascade (Eq. 16) fires identically: neural $\Gamma^2_n\uparrow$ spills into vascular, immune, metabolic, and renal channels. This *integral-path equivalence* unifies classic PTSD, generalised anxiety, and burnout under one D_Z threshold—only the time-domain shape of $\Gamma^2(t)$ differs.

Remark 4 (Disease onset as impedance phase transition)**.** *The positive-feedback term* $\beta_k\,\Gamma^4_k$ *in Eq. (16) creates a saddle-node bifurcation: below a critical forcing level,* Γ^2_k *remains at a low, stable equilibrium; above it, no low-*Γ *equilibrium exists and the channel collapses toward* $\Gamma^2_k = 1$*. Disease onset is therefore not a gradual deterioration but a* phase transition*—the system crosses*

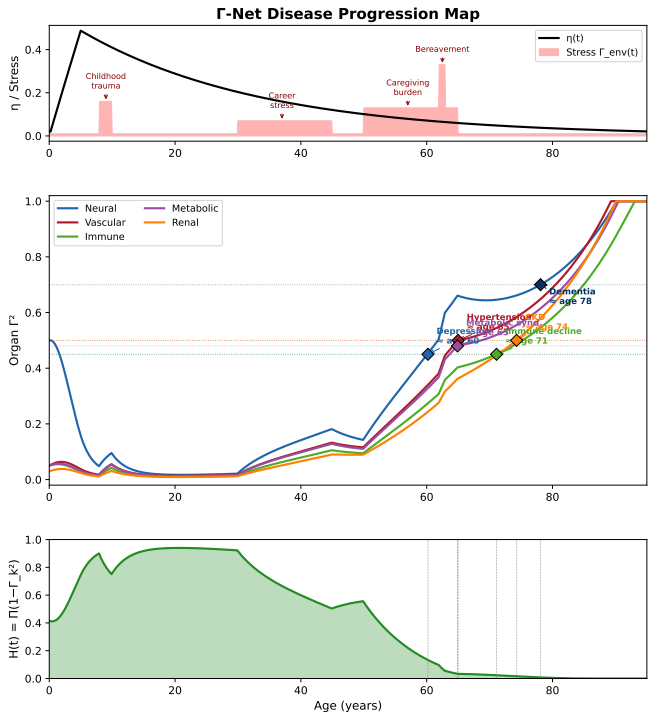

FIG. 1. Γ-Net disease progression map for a representative high-risk trajectory. **Top**: homeostatic repair capacity $\eta(t)$ and external stress profile. **Middle**: five coupled organ Γ^2 trajectories; diamonds mark sustained threshold crossings (disease onset). **Bottom**: total health product $H(t) = \prod_k (1 - \Gamma_k^2)$. All parameters are derived from Papers 2–3 physics; no curve fitting to clinical data.

a tipping point beyond which homeostatic repair cannot restore the low-Γ state. The coupling matrix C_{kj} then propagates this collapse to downstream organs, explaining why chronic diseases rarely appear in isolation.

D. Recovery dynamics: what happens when the stressor disappears

The preceding analysis raises the central clinical question: if the external stressor is removed ($s(t) = 0$ for $t > t_{\text{off}}$), does the system return to health? Setting $s = 0$ in Eq. (16) gives

$$\left.\frac{d\Gamma_k^2}{dt}\right|_{s=0} = -\eta(t)\,\Gamma_k^2 + \sum_{j\neq k} C_{kj}\,\Gamma_j^2 + \beta_k\,\Gamma_k^4 + \delta_k(t)\,. \quad (17)$$

Recovery requires the repair term $\eta\,\Gamma_k^2$ to dominate the sum of inter-channel coupling, positive feedback, and aging damage. Three regimes emerge:

1. **Pre-bifurcation (reversible).** If every channel satisfies $\Gamma_k^2 < \Gamma_k^{*2}$ (below the saddle-node threshold of Remark 4), then η exceeds the combined C_{kj} and β_k terms, and $\Gamma_k^2 \to \Gamma_{k,\text{eq}}^{\text{low}}$ exponentially. Clinically: acute stress, adjustment disorder, early burnout—symptoms resolve within weeks to months after stressor removal.

2. **Post-bifurcation, single channel (partially reversible).** If one channel (typically neural) has crossed its bifurcation point $\Gamma_n^2 > \Gamma_n^{*2}$ but the downstream channels have not, self-sustained positive feedback ($\beta_n\,\Gamma_n^4$) keeps the neural attractor locked even at $s = 0$. However, the coupling terms $C_{kn}\,\Gamma_n^2$ have not yet pushed other channels past their thresholds. Recovery requires *active intervention* to lower Γ_n^2 below the bifurcation point—pharmacotherapy (modulate η, widen basin), psychotherapy (provide corrective C2 gradient), or both (Paper 3, Sec. XIV.B). Once Γ_n^2 is pulled below Γ_n^{*2}, the remaining channels self-heal via their intact η. Clinically: GAD, moderate PTSD, early depression—treatable but not self-resolving.

3. **Post-bifurcation, multi-channel (structurally irreversible).** If the cascade has propagated through C_{kj} and two or more channels have independently crossed their bifurcation thresholds, each channel sustains the others even after the stressor is removed:

$$\eta\,\Gamma_k^2 < \sum_{j\neq k} C_{kj}\,\Gamma_j^2 + \beta_k\,\Gamma_k^4 \quad \forall\, k \in \text{locked channels}. \quad (18)$$

No single-channel intervention suffices because lowering Γ_k^2 in one channel is immediately re-driven by the locked neighbours. Clinically: Complex PTSD with metabolic syndrome, chronic pain with depression and immune decline—the "treatment-resistant" multimorbidity that consumes disproportionate healthcare resources. The only viable strategy is *simultaneous multi-channel intervention*: e.g. combined psychotherapy (neural) + exercise (vascular–metabolic) + anti-inflammatory (immune), each targeting a different Γ_k^2 below its respective bifurcation point.

Remark 5 (Why "just removing the stress" is sometimes not enough)**.** *The saddle-node bifurcation in Eq. (16) exhibits* hysteresis*: the stress level required to* trigger *the phase transition (s_{onset}) is lower than the stress* reduction *required to reverse it ($s_{recovery} < 0$, i.e. active repair is needed). In everyday language: "the damage is done" is not a metaphor but a statement about the bifurcation diagram. The gap between s_{onset} and $s_{recovery}$ widens with time spent in the high-Γ state, because aging damage $\delta_k(t)$ continues to accumulate and $\eta(t)$ continues to decline. Early intervention—before the cascade crosses a second channel's threshold—is therefore not merely preferable but* physically qualitatively different *from late intervention.*

The recovery hierarchy also predicts a *channel ordering* for rehabilitation: the channel with the lowest β_k/η ra-

tio recovers first when stress is removed, while the channel with the highest ratio is the last to unlock. In the five-organ model, vascular Γ_v^2 typically has the fastest self-repair (exercise increases η_v directly via shear-stress remodeling), while neural Γ_n^2 is slowest (structural Z_s changes require months of sustained C2 input). This predicts a clinically testable *recovery sequence*: physical health markers (blood pressure, HbA1c) improve before psychological symptoms (mood, sleep, cognition) — consistent with rehabilitation medicine observations that metabolic improvement often precedes and enables psychological recovery.

E. Tissue necrosis as thermodynamic substrate collapse

The recovery analysis above (Sec. IX D) assumes that the physical substrate—the material medium carrying impedance—remains intact. We now derive the condition under which the substrate itself is destroyed: *tissue necrosis*.

The impedance debt $D_Z = \int \Gamma^2 P_{\text{in}}\, dt$ is, by construction, the cumulative reflected energy that the tissue must either absorb or re-radiate. The repair machinery (C2: $\Delta Z = -\eta\, \Gamma\, x_{\text{in}}\, x_{\text{out}}$) converts a fraction of this energy into structural adaptation. When D_Z accumulates faster than C2 can process:

$$\frac{dD_Z}{dt} \;>\; \eta_{\max}\, |\Gamma|\, |x_{\text{in}}\, x_{\text{out}}|_{\max} \;\equiv\; \dot{D}^{\max}_{\text{repair}}\,, \qquad (19)$$

the excess energy has no impedance-remodeling pathway and is dissipated as heat and reactive biochemistry (free radicals, proteolytic enzymes, membrane disruption).

Definition 1 (Thermodynamic substrate collapse)**.** *Tissue necrosis is the state in which the material substrate Z_k loses the capacity for both transmission ($T = 0$) and remodeling ($\partial Z_k/\partial t = 0$ regardless of input). In the Γ-framework: $\Gamma_k = 1$* irreversibly*—not because the mismatch is large but because the medium that could change Z_k has been destroyed.*

A necrotic channel is a **topological black hole**: it reflects all incident energy, emits toxic degradation products (high-entropy noise) into the coupling matrix C_{kj}, and cannot be repaired by any endogenous C2 process. Unlike a high-Γ pathological attractor (Remark 4), which in principle can be reversed by sufficient intervention, a necrotic channel has undergone a *thermodynamic phase transition* from "mismatched but alive" to "structurally disintegrated."

The coupling drag on the remaining network is catastrophic. In Eq. (16), a necrotic channel contributes a constant $C_{kj} \times 1$ to every coupled neighbour, plus toxic noise that degrades neighbouring η_j. Left unaddressed, the necrotic source drives the global repair capacity $\eta(t)$ toward zero—the definition of *sepsis*: systemic exhaustion of the impedance-repair budget by a single overwhelming source of reflected energy and entropic waste.

1. Surgical intervention as topological truncation

When the necrotic channel satisfies Eq. (19) irreversibly and its coupling drag threatens systemic collapse, the only viable intervention is **topological truncation**: physically removing the necrotic node from the network.

Definition 2 (Topological truncation)**.** *Topological truncation is the application of* macroscopic external work *to sever all edges connecting a necrotic node to the network and to create a clean $\Gamma = 1$ boundary at the cut surface. In clinical practice: debridement, amputation, nephrectomy, limb-salvage resection.*

The physics of surgical truncation is:

1. **Remove the noise source.** The necrotic channel's toxic emissions ($C_{kj} \times 1$ coupling + entropic degradation of η_j) are eliminated from the ODE system.

2. **Create a clean boundary.** A surgically created $\Gamma = 1$ boundary (amputation stump, surgical margin) reflects 100% of incident energy but produces *no toxic noise*—it is an honest open circuit, not a necrotic one.

3. **Release hijacked bandwidth.** The massive nociceptive crosstalk (Paper 4, Sec. **??**) generated by the necrotic standing wave is terminated; the PFC's consciousness bandwidth is freed.

4. **Enable C2 recovery.** With the coupling drag removed, the global $\eta(t)$ stabilises, and surviving channels can resume repair per the recovery hierarchy of Sec. IX D.

Post-truncation, the brain's motor cortex faces a clean $\Gamma = 1$ boundary where the amputated limb was. If the cortex correctly detects this boundary and reduces $P_{\text{in}} \to 0$ (motor efference cessation), the reflected power vanishes and the network reaches a new, stable, albeit topologically reduced equilibrium. If the cortex continues to drive the severed channel—motor efference persists despite the open circuit—the reflected standing wave produces *phantom limb pain* (Paper 4, phantom_limb engine; Sec. **??**): the same bandwidth-hijacking mechanism, now at a surgically created rather than pathologically created $\Gamma = 1$ boundary.

Remark 6 (The therapeutic hierarchy of intervention)**.** *The Γ-framework orders clinical interventions by the severity of the impedance failure they address:*

Regime	***Intervention***	Γ ***mechanism***
Pre-bifurcation	*Remove stressor*	$s(t) \to 0$; η *restores*
Post-bifurcation (single)	*Pharmacotherapy + psychotherapy*	*Push* Γ_k^2 *below saddle-node*
Post-bifurcation (multi)	*Simultaneous multi-modal*	*Multi-channel* Γ_k^2 *reset*
Necrotic	*Topological truncation (surgery)*	*Remove node; clean* Γ=1 *boundary*

No intervention outside this hierarchy is physically meaningful: you cannot C2-remodel a substrate that no longer exists, and you cannot stressor-remove a bifurcation that is self-sustaining.

F. Caregiver triple-load model

Long-term caregivers experience three simultaneous Γ^2-injection channels with no comparable discharge path. Their impedance-debt rate is:

$$\frac{dD_Z}{dt} = \underbrace{\lambda\,\Gamma^2_{\text{other}}}_{\text{empathic load}} + \underbrace{\Gamma^2_{\text{moral}}}_{\text{moral constraint}} + \underbrace{\sigma\,s(t)}_{\text{daily stressor}} - \underbrace{\eta(t)\,\Gamma^2}_{\text{repair}}\,. \tag{20}$$

Channel 1: empathic load. Γ^2_{other} is the care-recipient's frozen mismatch (typically PTSD or dementia, $\Gamma^2 \approx 1$). The empathic coupling $\lambda > 0$ injects this energy into the caregiver's neural channel every contact hour; it cannot be "turned off" without severing the care relationship.

Channel 2: moral constraint. Social and internalised norms ("you must not give up") target the caregiver's fatigue pattern. Because the caregiver's η is already depleted, this pattern is approaching or inside $\mathcal{S}_{\text{eff}}$. By the Moral Constraint Theorem (Paper 4, Theorem **??**), the demand injects $\Gamma^2_{\text{moral}} > 0$ with $\Delta Z_{\text{corrective}} = 0$—pure injury.

Channel 3: moral prohibition on λ-reduction. The only physically available knob for reducing dD_Z/dt is to lower λ (set boundaries, take breaks, delegate). Moral pressure *forbids* this adjustment ("abandonment"), locking the intake valve open.

Remark 7 (Structural inevitability of caregiver disease)**.** *In Eq. (20), the first three terms are approximately constant or increasing (the care-recipient's Γ^2 does not spontaneously decrease; moral pressure does not spontaneously relent; daily stress persists). The repair term $\eta(t)\,\Gamma^2$ decreases because $\eta(t)$ declines by Arrhenius kinetics and is further depressed by cortisol-mediated hippocampal atrophy from the chronic load itself. Therefore $dD_Z/dt > 0$ for all future time—the deficit is* structurally guaranteed *to widen. The resulting disease cascade (Eq. 16, Fig. 1) is not a risk factor but a mathematical inevitability unless institutional intervention (shift rotation, λ-reduction protocols) is applied.*

Remark 8 (Compassion fatigue, burnout, vicarious trauma: one equation, three names)**.** *Clinical nomenclature distinguishes compassion fatigue, burnout, and vicarious trauma as separate syndromes. Within Eq. (20) they are three regimes of the same physics:*

Syndrome	*Dominant term*	$\mathcal{S}_{\text{eff}}$ *status*
Burnout	$\sigma\,s(t)$ *(workload)*	*not yet sealed*
Compassion fatigue	$\lambda\,\Gamma^2_{other}$	*approaching seal*
Vicarious trauma	$\lambda\,\Gamma^2_{other}$ *sealed into* $\mathcal{S}_{\text{eff}}$	*sealed*

*The physical distinction is whether the dominant Γ^2 pattern has entered $\mathcal{S}_{\text{eff}}$: if not, rest and workload reduction suffice (burnout); if sealed, only pharmacological η-elevation plus guided C2 gradient (combined therapy) can rewrite the pattern—identical to the PTSD treatment protocol (Paper 3, Sec. 13b.2). Rest alone cannot unseal the null space (Second Law, Paper 4, Theorem **??**).*

G. Surface observability: skin as impedance boundary

Long-term caregivers and high-empathy individuals sustain a chronic triple-load deficit (Eq. (20)). When $dD_Z/dt > 0$ persists, the accumulated impedance debt diffuses through the organ coupling matrix C_{kj}:

$$\text{Neural} \xrightarrow{C_{21}} \text{Vascular} \xrightarrow{C_{42}} \text{Metabolic} \to \text{Surface}$$

Neuralside Γ^2_n elevation first disrupts the hypothalamic–pituitary–adrenal (HPA) axis and autonomic regulation, raising vascular Γ^2_v; this in turn perturbs metabolic and immune C2 networks, ultimately manifesting as micro-circulatory and structural remodelling in peripheral tissue—especially skin and hair follicles. Crucially, the cascade passes through a *neuroendocrine–behavioural intermediate layer* before reaching the surface: disrupted sleep architecture, appetite dysregulation, irritability, and reduced exercise tolerance are all observable consequences of HPA-axis Γ^2 overload that temporally precede overt skin changes. The full observable chain is therefore

$$\underbrace{\text{Neural } \Gamma^2_n\uparrow}_{\text{mood, cognition}} \to \underbrace{\text{HPA / ANS}}_{\text{sleep, appetite, temper}} \to \underbrace{\text{Vascular } \Gamma^2_v\uparrow}_{\text{micro-circulation}} \to \underbrace{\text{Metaboli}}_{\text{collagen,}}$$

Remark 9 (Empathic overload as single-core running dual-core load)**.** *The empath's Γ-network physically simulates another person's pain state: her own daily mismatch load $\sigma\,s(t)$ remains, while the empathic channel adds $\lambda\,\Gamma^2_{other}$—effectively doubling the total Γ^2 input on a single repair budget $\eta(t)$. This is analogous to a single-core processor forced to sustain a dual-core workload: neither task can be suspended, thermal budget (η) is shared, and sustained over-clocking leads to monotonic performance degradation. The neuroendocrine–behavioural layer (HPA dysregulation, sleep fragmenta-*

tion, affect lability) constitutes the first externally observable "thermal throttling" of this overloaded system, preceding the dermal readout by weeks to months.

The critical observation is that skin is not an isolated organ but the *outermost boundary condition* of the whole-body impedance network. It simultaneously receives the output state of vascular, metabolic, and immune channels, making it a natural terminal load and real-time readout:

Internal Γ^2 channel	Skin manifestation	Colloquial term
Vascular $\Gamma^2_v \uparrow \rightarrow$ micro-circulatory mismatch	Pallor, dark circles, dull complexion	Bad colour
Metabolic $\Gamma^2_{\text{met}} \uparrow \rightarrow$ collagen & hydration imbalance	Roughness, sallowness, loss of elasticity	Poor skin quality
Immune $\Gamma^2_{\text{imm}} \uparrow \rightarrow$ inflammatory leakage	Redness, hypersensitivity, eczema flare-up	Skin deterioration

A third party's intuitive visual assessment—"You look worn out"—can therefore be re-read in the Γ-language as a low-resolution impedance tomography scan. The observer's cortex holds an internal template Z_{healthy} (hue, luminance, uniformity); when the observed surface impedance Z_{obs} deviates significantly, the visual reflection coefficient

$$\Gamma_{\text{visual}} = \frac{Z_{\text{obs}} - Z_{\text{healthy}}}{Z_{\text{obs}} + Z_{\text{healthy}}} \neq 0 \tag{21}$$

is read out in higher cortical areas as "something looks off."

Remark 10 (Detection threshold: null signal as health certificate)**.** *Equation (21) has an important converse. When diet, sleep, and stress are well regulated, all three load terms in Eq. (20) remain small:* $\sigma\, s(t) \approx 0$, $\lambda\, \Gamma^2_{other} \approx 0$, $\Gamma^2_{moral} \approx 0$. *Hence* $dD_Z/dt \lesssim 0$ *and the repair capacity* $\eta(t)$ *suffices to remodel daily mismatch back to baseline. Micro-circulation, collagen turnover, and immune tone stay within normal fluctuation—the skin terminal load is never pulled far from the healthy template, so* $Z_{obs} \approx Z_{healthy}$ *and* $\Gamma_{visual} \approx 0$. *The observer's perceptual read-out is therefore a* null signal*: "nothing looks wrong."*

This null signal is not an absence of information but a positive measurement: it certifies that all upstream Γ^2 *channels lie within the C2-repairable band. Conversely, once any channel sustains* Γ^2 *above the C2-repair ceiling—whether from chronic stress* ($\sigma\, s(t)$) *or empathic overload* ($\lambda\, \Gamma^2_{other}$)*—the surface deviation grows monotonically and the observer transitions abruptly from null signal to strong alarm ("this person has completely deteriorated"). The sharpness of this perceptual transition reflects the threshold nonlinearity inherent in the coupling cascade: sub-threshold* Γ^2 *is invisible; supra-threshold* Γ^2 *is unmistakable.*

Remark 11 (Downstream convergence of empathic and stress pathology)**.** *Empathic load* ($\lambda\Gamma^2_{other}$ *dominant) and occupational stress* ($\sigma\, s(t)$ *dominant) differ only in their* upstream *input channel. Once* Γ^2 *enters the coupling matrix* C_{kj}*, the subsequent path (Neural* $\rightarrow$ *Vascular* $\rightarrow$ *Metabolic* $\rightarrow$ *Surface) is identical. Consequently, from the skin's perspective as terminal load they are branches of the same* Γ*-disease tree (Sec. IX C), explaining why a caregiver who has tended a dementia patient for years and an executive under chronic workplace stress are visually indistinguishable in their surface presentation.*

H. Obesity as impedance-driven metabolic strategy

Conventional discourse frames obesity as a failure of self-control. The Γ-framework reveals a different causal structure: chronic Γ^2 overload triggers a whole-body metabolic reallocation that makes overeating a *physically rational* compensatory strategy rather than a moral deficiency.

a. Selfish-brain cascade. Under sustained stress ($\sigma\, s(t)$ large) or empathic overload ($\lambda\, \Gamma^2_{\text{other}}$ large), cumulative impedance debt D_Z rises and the neural channel reads persistent supply instability. The brain's response—well documented as the "selfish brain" hypothesis [35], which in Γ-Net topology is strictly an *asymmetric energy sink*: the brain occupies a privileged topological hub whose low effective impedance draws disproportionate power flux by Kirchhoff's current law—is to up-regulate caloric intake, particularly fast-availability substrates (sugar, fat), and to down-regulate satiety sensitivity. In the Γ-language:

$$\text{High } \Gamma^2_n \xrightarrow{\text{HPA axis}} \text{insulin/glucose redistribution} \xrightarrow{\text{reward}} \text{appetite}\uparrow \tag{22}$$

The behavioural output—compulsive eating—is the neural channel's attempt to stabilise its own D_Z at the expense of the metabolic channel.

b. Leptin resistance as impedance mismatch. In sustained obesity, circulating leptin is high, but the hypothalamic leptin-receptor pathway exhibits reduced signal transmission [37]—a textbook impedance mismatch. The satiety signal Z_{leptin} is present at the source but cannot couple into the load ($Z_{\text{hypothalamus}}$), so the brain never receives the "fat stores are sufficient" message. Formally, the leptin channel's reflection coefficient

$$\Gamma_{\text{leptin}} = \frac{Z_{\text{hypothalamus}} - Z_{\text{leptin}}}{Z_{\text{hypothalamus}} + Z_{\text{leptin}}} \rightarrow 1$$

approaches unity: the signal is almost totally reflected, and the transmitted fraction $T_{\text{leptin}} = 1 - \Gamma^2_{\text{leptin}} \rightarrow 0$. This is not a character flaw; it is a hardware-level C2-pathway failure.

c. Hedonic eating as short-term Γ^2 *reduction.* Ghrelin, leptin, and the mesolimbic reward circuit jointly modulate both hunger and hedonic valuation [36]. High-calorie intake produces a transient subjective Γ^2 reduction ("comfort"), trading a brief mismatch relief for a larger long-term D_Z increment (weight gain, insulin re-

sistance, cardiovascular load):

$$\underbrace{\Delta\Gamma^2_{\text{short}} < 0}_{\text{hedonic relief}}\,, \quad \underbrace{\Delta D_{Z,\text{long}} \gg 0}_{\text{metabolic debt}}$$

This is the metabolic analogue of the PTSD flashback–avoidance cycle (Paper 3): a locally rational response that is globally destructive. The dormant-to-explosive SNR transition (Paper 4, Eq. ptsd-snr) applies identically: quiet periods lower P_{noise}, making even weak cravings breach θ_{recall}.

Remark 12 (Moral blame as secondary metabolic injury)**.** *Applying the Moral Constraint Theorem (Paper 4, Theorem* **??***) to the obese individual: the eating pattern is already approaching or inside $\mathcal{S}_{eff}$ (sealed by years of reinforced reward circuitry), so moral condemnation ("lack of self-discipline") injects Γ^2_{blame} without any corresponding $\Delta Z_{corrective}$. The result is strictly additional impedance debt: the subject now carries the original metabolic* Γ^2 plus *a shame/stigma* Γ^2 *that further activates the stress–appetite cascade of Eq. (22), accelerating the very pathology it purports to correct. Weight-stigma research confirms precisely this paradox: moral pressure to lose weight predicts* increased *weight gain over time [38].*

Remark 13 (Positive-feedback closure: obesity as mathematical attractor)**.** *The metabolic cascade of Eq. (22) is not a linear chain but a* closed positive-feedback loop*:*

$$\textit{Metabolic } \Gamma^2\!\uparrow \rightarrow \textit{ peripheral deterioration } \rightarrow \textit{ neural detection } \rightarrow \textit{ energy reallocation } \rightarrow \textit{ appetite}\!\uparrow \rightarrow \textit{ Metabolic } \Gamma^2\!\uparrow$$

Each cycle amplifies the previous round's Γ^2*. As long as the environmental stress $\sigma\, s(t)$ persists, the loop gain exceeds unity and cumulative weight gain is not a lifestyle choice but a* fixed-point attractor *of the coupled dynamics—obesity emerges as a mathematical inevitability on physiological timescales.*

Remark 14 ("Willpower" as $\eta_{\text{pfc}}(t)$: a depleting physical quantity, not a moral virtue)**.** *What common discourse calls "willpower" or "self-control"—in control-theoretic terms, a* top-down prefrontal forcing function *that attempts to override subcortical drive signals—corresponds, in the* Γ*-language, to the prefrontal repair rate $\eta_{pfc}(t)$. But the prefrontal cortex is itself a channel within the same impedance network; its $\eta_{pfc}(t)$ decays by the same Arrhenius kinetics and is further depressed by the chronic cortisol load generated by the metabolic–stress cascade. Therefore $\eta_{pfc}(t)$ is not a free parameter that can be summoned at will; it is a* state variable *already depleted by the very process it is asked to override:*

$$\eta_{pfc}(t) = \eta_0\, e^{-E_a/k_B T}\cdot f_{cortisol}(D_Z(t)) \xrightarrow[\textit{chronic } \Gamma^2]{} \textit{ depleted}$$

Asking a chronically stressed or obese individual to "simply exercise self-control" is therefore not a moral appeal but a request to draw on a resource that the physics has already consumed. The illusion that willpower is a character trait rather than a depletable physical budget arises because, in low-Γ^2 *individuals (low stress, intact* η*), prefrontal override appears effortless—just as a lightly loaded CPU appears to have "infinite patience." The concept dissolves once the load exceeds the repair capacity, which is precisely the regime in which it is most demanded.*

Proposition 2 (Non-existence of free willpower: proof by contradiction from psychiatric epidemiology)**.** Claim. *"Willpower" is not a free parameter of the human system.*

Proof. *Suppose, for contradiction, that willpower were a freely available override—i.e. that every agent possesses an unbounded η_{pfc} capable of suppressing any* Γ^2 *pattern at any time. Then no mismatch could persist against the agent's intention: every depressive episode could be willed away, every compulsive behaviour halted, every anxiety loop interrupted. In other words,* Γ^2 *could never accumulate to clinical threshold in any channel, and* no mental illness could exist*.*

Empirically, approximately 970 million people worldwide carry a diagnosed mental disorder as of 2019 (WHO Global Burden of Disease). This is a direct contradiction. Therefore the premise is false: willpower is not a free parameter but a state-dependent, depletable physical quantity $\eta_{pfc}(t)$, subject to the same Arrhenius decay, cortisol suppression, and Γ^2*-induced degradation as every other channel in the impedance network.* □

Paper 4 [5] refines this further: even when η_{pfc} is metabolically adequate, effective willpower requires (i) trained thalamic gate speed (Q_{gate}) and (ii) a calibrated impedance map ($1 - \langle\Gamma^2_{\text{appetite}}\rangle$); the three factors compose multiplicatively (Paper 4, Eq. (14) [5]), so a single zero factor extinguishes volitional capacity regardless of the other two.

I. Immune–metabolic bidirectional feedback: the inflammation–insulin trap

Among the off-diagonal entries of the coupling matrix C_{kj} in Eq. (16), the pair ($C_{\text{imm,met}}$, $C_{\text{met,imm}}$) deserves independent analysis because *both* are positive and mutually reinforcing, creating a local positive-feedback subsystem even when the remaining channels are stable.

Write the two-channel subsystem:

$$\frac{d\Gamma^2_{\text{imm}}}{dt} = -\eta\,\Gamma^2_{\text{imm}} + C_{\text{imm,met}}\,\Gamma^2_{\text{met}} + \beta_{\text{imm}}\,\Gamma^4_{\text{imm}}\,, \quad (23)$$

$$\frac{d\Gamma^2_{\text{met}}}{dt} = -\eta\,\Gamma^2_{\text{met}} + C_{\text{met,imm}}\,\Gamma^2_{\text{imm}} + \beta_{\text{met}}\,\Gamma^4_{\text{met}}\,. \quad (24)$$

The mutual-excitation terms form a closed cycle: adipocyte hypertrophy raises metabolic Γ^2_{met} (insulin resistance, hepatic lipogenesis); this triggers macrophage infiltration and pro-inflammatory cytokine release (IL-6, TNF-α), raising immune Γ^2_{imm} (chronic low-grade inflammation, elevated CRP); the inflammatory milieu further impairs insulin receptor signalling, pushing Γ^2_{met} higher still:

$$\Gamma^2_{\text{met}}\!\uparrow \xrightarrow{C_{\text{imm,met}}} \Gamma^2_{\text{imm}}\!\uparrow \xrightarrow{C_{\text{met,imm}}} \Gamma^2_{\text{met}}\!\uparrow \rightarrow \cdots$$

Linearising about a small perturbation $\delta\Gamma_k^2$ around equilibrium, the Jacobian of the subsystem has eigenvalues

$$\lambda_\pm = -\eta + \tfrac{1}{2}(\alpha_{\rm imm}+\alpha_{\rm met}) \pm \sqrt{C_{\rm imm,met}\, C_{\rm met,imm} + \tfrac{1}{4}(\alpha_{\rm imm}-\alpha_{\rm met})^2}\,, \tag{25}$$

where $\alpha_k \equiv 2\beta_k\,\Gamma_{k,\rm eq}^2$. The subsystem destabilises when $\lambda_+ > 0$, i.e.

$$\sqrt{C_{\rm imm,met}\, C_{\rm met,imm}} > \eta - \tfrac{1}{2}(\alpha_{\rm imm}+\alpha_{\rm met})\,. \tag{26}$$

This is the *inflammation–insulin trap*: once the geometric mean of the mutual coupling exceeds the net repair margin, the two channels lock into a self-sustaining high-Γ^2 fixed point—metabolic syndrome with chronic inflammation—even if the original neural stressor has been removed.

Clinical correlates are well documented: obesity → elevated CRP → type 2 diabetes → further weight gain constitutes the "metabolic–inflammatory" comorbidity cluster (*coupled-subsystem divergence* in Γ language: two channels simultaneously locked above their respective θ). In recovery-dynamics terms (Sec. IX D), this sub-loop can achieve regime 3 (structurally irreversible) *independently* of the neural channel, explaining why bariatric surgery (which mechanically breaks the metabolic $\beta_{\rm met}\,\Gamma^4$ term) can resolve both diabetes *and* systemic inflammation simultaneously—a dual-channel reset that diet alone, targeting only $\Gamma_{\rm met}^2$, often cannot achieve.

J. Type 2 diabetes as global impedance drift: from receptor mismatch to six-channel deadlock

The preceding subsections established the motivational cascade (Sec. IX H) and the immune–metabolic trap (Sec. IX I). We now derive the complete physics of type 2 diabetes mellitus (T2DM) from the three constraints alone, as the first non-neural disease model carried to the same depth as the PTSD three-layer mechanism (Paper 4, Sec. 7.10).

1. *Insulin signalling as impedance matching*

The insulin–glucose regulatory loop is a *negative feedback impedance servo*. Pancreatic β-cells secrete insulin (the "source") at impedance Z_β; peripheral cells (muscle, liver, adipose) present insulin receptors (the "load") at impedance Z_R. Under healthy matching, $\Gamma_{\rm ins} = (Z_R - Z_\beta)/(Z_R + Z_\beta) \approx 0$ and the transmission coefficient $T_{\rm ins} = 1 - \Gamma_{\rm ins}^2 \approx 1$: nearly all insulin signal is absorbed by the receptor, glucose is cleared, and the loop is quiescent.

Define the *receptor impedance* in terms of receptor density N_R and binding affinity K_d:

$$Z_R = \frac{Z_0}{K_d \cdot N_R}\,, \tag{27}$$

where Z_0 is a tissue-type normalisation constant (Table **??**, row 9). Insulin resistance is *not* a binary switch but a continuous drift: either $N_R \downarrow$ (receptor downregulation) or $K_d \uparrow$ (binding affinity loss) raises Z_R, increasing $\Gamma_{\rm ins}^2$ monotonically.

2. *C2 remodeling of receptor impedance: the global baseline drift*

Unlike neural channels (where Z changes along specific axonal pathways), the insulin–receptor system operates as a *chemical bath*: insulin is broadcast systemically, and every cell's Z_R is exposed to the same hormonal milieu. A chronic stressor (elevated cortisol via the HPA axis, Eq. 22) acts as a *global* $x_{\rm in}$ that drives C2 remodeling on every peripheral receptor simultaneously:

$$\frac{dZ_R}{dt} = \underbrace{-\eta_R\,\Gamma_{\rm ins}\,x_{\rm ins}\,x_{\rm glucose}}_{\text{C2 repair (toward } Z_\beta)} + \underbrace{\alpha_c\,\Gamma_{\rm cortisol}^2\,Z_R}_{\text{cortisol-driven upregulation}}\,, \tag{28}$$

where η_R is the receptor repair rate (analogous to neural η), $x_{\rm ins}$ and $x_{\rm glucose}$ are the insulin and glucose signals at the receptor, and α_c is the cortisol coupling coefficient. The first term is standard C2: it drives Z_R toward Z_β (healthy matching). The second term is the endocrine-specific physics: chronically elevated cortisol ($\Gamma_{\rm cortisol}^2 \gg 0$) acts as a *counter-gradient* that pushes Z_R upward faster than η_R can pull it down.

This is the fundamental difference between neural pathology and metabolic pathology. In the neural case, C2 scars form on *individual pathways* (point-to-point high-Γ channels). In the metabolic case, C2 remodeling acts on *all receptors globally*—a low-frequency, whole-body impedance drift that is physically equivalent to changing the dielectric constant of the entire transmission-line medium.

3. *The insulin resistance lock: bifurcation analysis*

Setting $dZ_R/dt = 0$ in Eq. (28) and solving for the equilibrium $\Gamma_{\rm ins}$:

$$\Gamma_{\rm ins}^* = \frac{\alpha_c\,\Gamma_{\rm cortisol}^2\,Z_R^*}{\eta_R\,x_{\rm ins}\,x_{\rm glucose}}\,. \tag{29}$$

When $\Gamma_{\rm cortisol}^2$ is small (low stress), $\Gamma_{\rm ins}^* \approx 0$ and the servo holds. As stress chronically elevates cortisol, the equilibrium $\Gamma_{\rm ins}^*$ drifts toward 1.

The critical regime is the *positive feedback threshold*. As $\Gamma_{\rm ins}^2$ rises, less glucose is cleared, hyperglycaemia develops, and the pancreas compensates by secreting more insulin. But elevated insulin itself drives receptor downregulation ($N_R \downarrow$), further raising Z_R:

$$\frac{dN_R}{dt} = -\kappa\,[\text{insulin}]\,N_R + \sigma_R\,(N_{R,0} - N_R)\,, \tag{30}$$

where κ is the internalisation rate and σ_R is the receptor synthesis rate. Substituting into Eq. (27), the feedback loop closes:

$$\Gamma^2_{\text{ins}}\uparrow \xrightarrow{\text{hyperglycaemia}} \text{insulin}\uparrow \xrightarrow{\kappa} N_R\downarrow \xrightarrow{Z_R\uparrow} \Gamma^2_{\text{ins}}\uparrow \rightarrow \cdots$$

This is structurally identical to the immune–metabolic trap of Eqs. (23)–(24), but occurring *within* the endocrine channel. The bifurcation condition is:

$$\kappa\,[\text{insulin}]\,\frac{\partial Z_R}{\partial N_R} > \eta_R \quad \Longleftrightarrow \quad \frac{\kappa\,[\text{insulin}]}{K_d\,N_R^2} > \frac{\eta_R}{Z_0}\,. \tag{31}$$

Once the product of hyperinsulinaemia and receptor internalisation rate exceeds the repair margin, the system is locked in a high-Γ^2_{ins} attractor—*insulin resistance is structurally irreversible under the current* η_R, precisely analogous to the PTSD scar being locked in $\mathcal{S}_{\text{eff}}$.

Remark 15 (Why diet alone often fails)**.** *Caloric restriction reduces $x_{glucose}$, which weakens the C2 repair term in Eq. (28) by lowering the product $x_{ins}\cdot x_{glucose}$. If the cortisol term $\alpha_c\,\Gamma^2_{cortisol}$ remains elevated (chronic stress persists), the net dZ_R/dt may remain positive* despite caloric deficit*—the receptor impedance continues to drift upward even as the patient loses weight. This resolves the clinical paradox of patients who achieve significant weight loss yet remain insulin-resistant: the* endocrine *channel (Z_R drift) was never addressed because the intervention targeted only the* metabolic *channel (Γ^2_{met}).*

4. *The six-channel T2DM cascade*

Once Γ^2_{ins} crosses the bifurcation threshold of Eq. (31), the coupled dynamics of Eq. (16) propagate the mismatch through the full coupling matrix C_{kj}. We trace the complete cascade through six channels:

1. **Neural → Endocrine** ($C_{\text{end,neu}}$): Chronic stress raises $\Gamma^2_{\text{cortisol}}$ via the HPA axis. This is the *initiating event*—the "match that lights the fire." The cortisol signal acts as a global x_{in} that drives receptor Z_R upward (Eq. 28).

2. **Endocrine → Metabolic** ($C_{\text{met,end}}$): As $\Gamma^2_{\text{ins}} \to 1$, glucose clearance collapses ($T_{\text{ins}} \to 0$). Hyperglycaemia and hyperinsulinaemia both rise; hepatic lipogenesis accelerates (the liver converts excess glucose to fat when the metabolic channel cannot absorb it). Γ^2_{met} rises.

3. **Metabolic → Immune** ($C_{\text{imm,met}}$): Adipocyte hypertrophy from lipogenesis triggers macrophage infiltration and pro-inflammatory cytokine release (IL-6, TNF-α). Γ^2_{imm} rises—the inflammation–insulin trap of Sec. IX I activates.

4. **Immune → Vascular** ($C_{\text{vas,imm}}$): Chronic inflammation damages endothelial function. The vascular channel's Γ^2 rises: arterial compliance drops per the Moens-Korteweg equation $c = \sqrt{Eh/2\rho r}$, where increased wall thickness h (fibrosis) and decreased radius r (lumen narrowing) both elevate pulse-wave velocity—the vascular analogue of rising Z. This is the physics of diabetic macrovascular disease.

5. **Vascular → Renal** ($C_{\text{ren,vas}}$): Elevated vascular Γ^2 reduces glomerular perfusion. The renal channel compensates by activating RAAS (renin–angiotensin–aldosterone), further raising systemic vascular Z—a second positive-feedback loop. Proteinuria and declining GFR are the impedance signatures of diabetic nephropathy.

6. **Renal → Neural** ($C_{\text{neu,ren}}$): Uraemic toxins from declining renal function impair neural repair rate η_n, further depressing the brain's ability to modulate the HPA axis—closing the six-channel loop back to channel 1.

The complete cascade forms a closed hexagonal coupling:

$$\text{Neu} \xrightarrow{C_{\text{end,neu}}} \text{End} \xrightarrow{C_{\text{met,end}}} \text{Met} \xrightarrow{C_{\text{imm,met}}} \text{Imm} \xrightarrow{C_{\text{vas,imm}}} \text{Vas} \xrightarrow{C_{\text{ren,vas}}} \text{Ren} \tag{32}$$

Once any *two* adjacent channels are locked above their respective bifurcation thresholds (regime 3 of Sec. IX D), single-channel intervention is provably insufficient:

$$\eta_k\,\Gamma^2_k < \sum_{j\neq k} C_{kj}\,\Gamma^2_j + \beta_k\,\Gamma^4_k \quad \forall\, k \in \text{locked channels}. \tag{33}$$

This is Eq. (18) instantiated for the T2DM hexagon. The clinical implication is immediate: treating only blood sugar (metformin targeting Γ^2_{met}) without simultaneously addressing stress ($\Gamma^2_{\text{cortisol}}$), inflammation ($\Gamma^2_{\text{imm}}$), and vascular damage ($\Gamma^2_{\text{vas}}$) leaves the locked channels to redrive Γ^2_{ins} from the other side of the hexagon.

Theorem 2 (T2DM treatment minimum: multi-channel simultaneous intervention)**.** *Let $\mathcal{L} \subseteq \{1,\ldots,6\}$ be the set of channels that have individually crossed their bifurcation thresholds in the T2DM hexagon (Eq. 32). A therapeutic protocol can reduce Γ^2_k below threshold for all $k \in \mathcal{L}$ only if it targets at least $|\mathcal{L}|$ channels simultaneously.*

Proof. *By Eq. (33), each locked channel k receives drive from its neighbours that exceeds its own repair capacity. Unlocking channel k alone reduces Γ^2_k transiently, but the coupling $C_{kj}\,\Gamma^2_j$ from the remaining locked channels $j \in \mathcal{L}\setminus\{k\}$ re-elevates Γ^2_k above threshold within $\tau_{relapse} \sim C_{kj}/\eta_k$ ticks. Only simultaneous reduction of all Γ^2_j for $j \in \mathcal{L}$ breaks the mutual sustenance condition.* □

This theorem explains the well-documented clinical observation that T2DM management improves dramatically under "lifestyle medicine" protocols that combine dietary change (metabolic), exercise (vascular +

metabolic), stress reduction (neural + endocrine), and anti-inflammatory agents (immune)—each targeting a different vertex of the hexagon. It also predicts that GLP-1 receptor agonists (which simultaneously reduce appetite, lower $\Gamma^2_{\rm met}$, and exhibit anti-inflammatory properties) should outperform agents that target only one channel—a prediction consistent with the clinical superiority of semaglutide over metformin monotherapy.

Remark 16 (β-cell exhaustion as impedance substrate collapse)**.** *In advanced T2DM, the compensatory hyperinsulinaemia of stage 2 eventually exhausts β-cell secretory capacity: $Z_\beta \to \infty$ (cell death). This is the metabolic analogue of tissue necrosis (Paper 5, Sec. IX E): the substrate that* generates *the source impedance is destroyed, and no amount of C2 remodeling can restore a channel whose source no longer exists. At this stage, T2DM functionally converges with T1DM—not because the aetiology is the same, but because the* topological endpoint *is identical: $Z_\beta = \infty$, $\Gamma_{ins} = 1$, $T_{ins} = 0$. Exogenous insulin injection is the only remaining intervention: it replaces the destroyed source by providing an external Z_β to the network.*

K. Acute versus chronic stress: bifurcation threshold under time-limited forcing

The stress-injection term $\sigma_k\, s(t)$ in Eq. (16) does not distinguish beneficial from harmful stress; the distinction emerges from the *temporal integral* relative to the repair term.

Consider a square-pulse stressor of amplitude A and duration Δt:

$$s(t) = A\left[\Theta(t) - \Theta(t-\Delta t)\right], \tag{34}$$

where Θ is the Heaviside step. During the pulse, the channel trajectories satisfy

$$\Gamma^2_k(t) \approx \Gamma_{k,\rm eq} + \frac{\sigma_k\, A}{\eta}\left(1 - e^{-\eta\, t}\right), \tag{35}$$

to first order (neglecting β_k and C_{kj} for short pulses). The peak excursion is bounded by $\Delta\Gamma^2_k \le \sigma_k A/\eta$, and after the pulse ends, η drives the system back exponentially with time constant $1/\eta$. The system remains safe (regime 1 of Sec. IX D) provided

$$\Gamma_{k,\rm eq} + \frac{\sigma_k\, A}{\eta} < \Gamma^{*2}_k \quad \text{(bifurcation threshold)}. \tag{36}$$

This is the physics of *beneficial acute stress*: high-intensity exercise (A large, Δt small), cold-water immersion, intermittent fasting—all push Γ^2_k transiently upward, stimulating C2 repair ("hormesis"), and the system returns to baseline before coupling or positive feedback can engage.

Contrast this with chronic stress: $s(t) = a$ (constant, $a < A$) for $t \gg 1/\eta$. The steady-state solution of Eq. (16) now includes coupling:

$$\Gamma^2_{k,\rm ss} = \frac{\sigma_k\, a + \sum_j C_{kj}\, \Gamma^2_{j,\rm ss} + \delta_k}{\eta - \beta_k\, \Gamma^2_{k,\rm ss}}. \tag{37}$$

Even for small a, the denominator shrinks as $\Gamma^2_{k,\rm ss}$ rises (positive feedback β_k), and the coupling terms C_{kj} inject additional load from neighbouring channels. The saddle-node bifurcation occurs at

$$\eta = \beta_k\, \Gamma^{*2}_k + \frac{\partial}{\partial \Gamma^2_k} \sum_j C_{kj}\, \Gamma^2_j, \tag{38}$$

a threshold that *decreases* with age (because $\eta(t)$ declines by Arrhenius kinetics while $\delta_k(t)$ grows).

Remark 17 (The exercise paradox resolved)**.** *Exercise is simultaneously a high-Γ^2 perturbation (muscles, cardiovascular) and the most effective known intervention against chronic disease. The paradox dissolves once time scale is explicit: exercise is* acute *forcing ($\Delta t \sim 1$ hour, then $s \to 0$) that never reaches the bifurcation threshold (Eq. 36), while the transient Γ^2 elevation* stimulates *C2 remodeling—raising effective η in vascular and metabolic channels (shear-stress–mediated endothelial adaptation, mitochondrial biogenesis). Chronic sedentary stress, by contrast, is low-amplitude but* never turns off*: the system drifts toward the chronic steady state (Eq. 37) and eventually crosses the bifurcation. Exercise is therapeutic precisely because it is* intermittent and bounded.

X. E0 EMERGENCE: DISEASE FROM TOPOLOGY ALONE

The twenty-nine Γ-networks (Table III) and the central controllers establish C2 universality in the *scalar* domain (one impedance channel per tissue). A stronger claim follows: clinical pathology should *emerge* from the *multi-dimensional* topology of the Γ-network without any disease-specific code.

We define three emergence levels:

- **E0 (True Emergence)**: only C1/C2/C3 + initial conditions. Zero disease-specific flags, thresholds, or formulas.
- **E1 (Parameterised)**: C2 via `ImpedanceChannel.remodel()` with per-disease parameters ($Z_{\rm coeff}$, severity).
- **E2 (Scripted)**: ad-hoc Z updates. Legacy only; forbidden for new code.

E0 is the gold standard: "if you need a flag, it didn't emerge." We test this with 29 tissue blueprints (`GammaTopology` instances with biologically assigned `TissueType` and mode count K; Paper 0 [1], Table 4) and pure perturbations of initial conditions. Detailed disease-emergence experiments are shown below for the

cardiovascular (11 nodes) and neural (9 nodes) systems; all remaining 27 blueprints pass the same C1/C2 physics verification (245 parametrised tests, 100% pass).

A. Cardiovascular E0: hypertension, MI, aortic stenosis

Starting from a healthy 11-node cardiovascular `GammaTopology` ($K \in \{4, 2, 1\}$; Paper 1 [2], Sec. 4.5):

1. **Hypertension**: arteriolar $Z \times 2.0 \Rightarrow \Gamma^2$ at aorta→arterioles increases; downstream capillary transmission decreases; C2 partially compensates over 300 ticks (remodeling = physiological adaptation).
2. **Myocardial infarction**: sever aorta→coronary edge ⇒ coronary node isolated; total transmission decreases; topology is structurally different from HTN (correlation < 0.99).
3. **Aortic stenosis**: valve $Z \times 3.0 \Rightarrow$ LV-to-aorta Γ^2 increases (obstruction physics).
4. **Dose–response**: arteriolar Z-factor $\{1.0, 1.5, 2.0, 3.0\}$ yields monotonically increasing Γ^2—severity scales with perturbation magnitude.

Thirteen automated tests (`tests/test_e0_cardiovascular.py`), 100% pass. No disease model was coded; pathology identity is determined solely by the perturbation type (impedance shift vs. edge removal) and magnitude.

B. Neural E0: ALS, stroke, PTSD, phantom pain

Starting from a healthy 9-node neural `GammaTopology` ($K \in \{5, 3, 2, 1\}$; Paper 1 [2], Sec. 4.6):

1. **ALS** (motor neuron $Z \times 3.0$): Γ^2 at spinal→motor edge increases; cumulative A_{imp} over 30 ticks is 2.9× healthy; C2 partially compensates (plasticity).
2. **Stroke** (cortex→thalamus edge severed): descending motor transmission interrupted; ascending sensory pathway spared (different edges)—the deficit pattern is topologically determined.
3. **PTSD** (nociceptor C driven at 5× normal amplitude for 100 ticks, then stopped): pain-pathway Γ^2 is altered during trauma *and* a residual mismatch ($\varepsilon \approx 1$) persists after stimulus removal, locked in $\mathcal{S}_{\text{eff}}$. The recalled intensity depends on $\text{SNR}_{\text{recall}} = \varepsilon^2 P_{\text{probe}}/P_{\text{noise}}$ (Paper 4, Theorem recall-snr): quiet environments lower P_{noise}, causing submerged traces to breach θ_{recall} explosively—the "flashback" is a dormant-to-explosive SNR transition, not a static replay.
4. **Phantom limb pain** (motor neuron→nociceptor edge severed): efferent modulation lost; nociceptor Γ^2 altered; ascending pain pathway remains intact—phantom "sensation" has a physical substrate (open-circuit reflection).

Seventeen automated tests (`tests/test_e0_neural.py`), 100% pass. Additional tests verify:

- ALS vs. stroke produce distinguishable Γ-trajectories (correlation < 0.99).
- Severity dose–response is monotonic (early ticks, before C2 convergence).
- $A_{\text{cut}} > 0$ at all times (irreducibility, K=5 → 3 → 2 → 1).
- A_{cut} remains > 0 after 300 ticks of C2 evolution (irreducible, not learnable).
- A_{imp} decreases under C2 (learnable).
- Neural $A_{\text{cut}} \geq$ cardiovascular $A_{\text{cut}} \times 0.5$ (more K-level transitions ⇒ higher geometric cost).

C. Implications

The 275 E0 tests (245 parametrised blueprint tests + 13 cardiovascular + 17 neural disease tests) across all 29 organ-system topologies prove that clinical pathology *can* emerge from pure impedance physics—C1/C2/C3 plus initial conditions—on biologically realistic tissue topologies. No disease-specific code, no flags, no thresholds, no scoring systems. The disease *is* the perturbation; the clinical pattern *is* the resulting Γ-distribution.

D. Thermal loop collapse: hypothermia and hyperthermia

Paper 3 [4] derives the $\Gamma^2 \leftrightarrow T$ self-consistent loop: impedance mismatch produces heat, heat maintains temperature, temperature sets viscosity and repair rate, which in turn set Γ. The 37 °C operating point is the stable fixed point of this loop. When the loop is driven outside its basin of attraction, two E0 pathologies emerge—requiring no disease-specific code:

	Hypothermia	**Hyperthermia**
Direction	$T \ll T^*$	$T \gg T^*$
Primary driver	$\eta(T)\uparrow$ (viscosity)	$k_B T\uparrow$ (noise)
Γ mechanism	$Z_{\text{bus}}\uparrow \rightarrow \Gamma^2\uparrow$	$\delta\Gamma/\Gamma \propto K^{3/2}\delta T/T$
First failure	Global Γ^2 rise	PFC (K=5) first
Clinical sign	Bradycardia → coma	Delirium → seizure
Feedback	Positive (less heat → colder)	Positive (more noise –
C2 repair	Slowed (Arrhenius ↓)	Overwhelmed (noise >

Both are **E0**: they arise from C1/C2/C3 plus the Arrhenius temperature dependence of Z_{bus} and η_{learn}—no pathology-specific parameters. The clinical asymmetry (hypothermia is global; hyperthermia hits PFC first) is a direct consequence of the $K^{3/2}$ thermal vulnerability derived in Paper 4 [5].

This does not mean all 117 disease models in the codebase are E0 today—most remain E1 (15) or legacy E2 (102). But the existence proof changes the target: E0 is achievable, and every E2 model is a candidate for elimination via topology-based emergence.

E. Sleep disorders as gate-leak pathology

Paper 3 [4] derives the gate-leak inequality:

$$G_i^2\,\Gamma_i^2\,P_{\text{in},i} \;>\; 4\,k_B T\,R_{\text{eff}}\,\Delta f \;\equiv\; P_\theta \qquad \text{(P3, Eq. 10)}$$

as the physical condition for forced awakening. Three clinically distinct sleep disorders map onto three different structural failures within this single inequality:

a. Insomnia: Γ_i^2 overload. Chronic stress or trauma forges one or more channels at high structural mismatch ($\Gamma_i^2 \gg \langle\Gamma^2\rangle$). The thalamic gate G_i is at its physical minimum, yet the leaked power still exceeds P_θ. Clinical consequence: the patient "cannot fall asleep" despite exhaustion. The standard epidemiological correlates—anxiety, rumination, chronic pain—are precisely the conditions that elevate Γ_i^2 on specific channels. Pharmacological intervention (e.g. benzodiazepines) temporarily increases gate attenuation (lowers G_i); durable resolution requires C2 remodelling of the underlying Γ_i^2 (cognitive-behavioural therapy for insomnia, CBT-I, provides the sustained counter-signal).

b. Narcolepsy: Z_{anchor} loss. The orexin (hypocretin) system provides the structural anchor Z_{anchor} that prevents Z_{arousal} from drifting below the consciousness threshold. Autoimmune destruction of orexin neurons (type-1 narcolepsy) eliminates this anchor; Z_{arousal} becomes a random walk that crosses threshold at unpredictable intervals → sleep attacks. Cataplexy—the sudden loss of muscle tone during strong emotion—is the motor-gate analogue: a limbic Γ^2 spike leaks through the descending motor gate ($G_{\text{motor}} \to 0$), the same gate-leak equation applied to the efferent rather than afferent pathway. Orexin receptor agonists (suvorexant class, inverse; future orexin-2 agonists for narcolepsy) directly target Z_{anchor}.

c. Hypersomnia: β_{recal} deficit. Idiopathic hypersomnia (IH) preserves the arousal anchor but has a structurally low NREM recalibration rate β_{recal}. Impedance debt D_Z is never fully discharged during sleep; the organism requires abnormally long sleep times and wakes with "sleep inertia" (residual D_Z). During the day, elevated baseline $\langle\Gamma^2\rangle$ means the gate-leak condition barely holds: any small increase in P_{in} (warm room, monotonous task) tips a channel past P_θ and the system collapses back into sleep → excessive daytime sleepiness. Wake-promoting agents (modafinil, solriamfetol) reduce the effective D_Z load by increasing P_{in} uniformly across channels, keeping $G_i^2\,\Gamma_i^2\,P_{\text{in}}$ above the "awake" operating regime.

d. Unified diagnostic map.

Disorder	Broken parameter	Gate-leak status	Treat
Insomnia	Γ_i^2 too high	Cannot close (leak $> P_\theta$)	Lower
Narcolepsy	$Z_{\text{anchor}} \to 0$	Collapses suddenly	Resto
Hypersomnia	β_{recal} low	Barely holding	Raise

All three disorders are *structural* impedance failures—each specifying a different physical parameter—unified under a single gate-leak equation. The clinical fact that insomnia and hypersomnia often coexist ("tired but wired") maps naturally to a system where β_{recal} is low (undischarged D_Z) *and* one channel has high Γ_i^2 (chronic leak): the patient is simultaneously too depleted to stay awake and too activated to fall asleep.

XI. EMPIRICAL EVIDENCE: NETWORK PROPAGATION

This paper (Section III) validated Γ within single organ systems. Here we test a stronger prediction: adding a physical measurement to one Γ-network should improve predictions in *coupled* networks through impedance propagation at inter-network interfaces.

A. Experimental design

Using NHANES 1999–2018 (10 cycles, n = 49,774, 7,770 deaths, up to 20-year follow-up), four configurations:

A Laboratory values only (27 items → 12 organs)

B Labs + blood pressure (SBP, DBP, pulse pressure → vascular, cardiac, renal, neurological Z)

C Labs + smoking status (→ pulmonary, vascular, immune Z)

D Full physics (Labs + BP + smoking)

All mappings use textbook reference intervals. **Zero parameters are fitted to any outcome data.**

B. Results

Cardiac death AUC improves from 0.508 (chance) to 0.674—$\Delta = +0.166$—from a vascular measurement, not a cardiac measurement. SBP physically propagates through the vascular–cardiac interface (afterload), so vascular Z directly determines cardiac Z.

Endocrine serves as a negative control: no significant vascular–endocrine pressure interface exists, and AUC is unchanged ($\Delta = 0.000$).

TABLE IV. All-cause mortality AUC by configuration.

Configuration	AUC (95% CI)	Δ vs. A
A: Labs only	0.629 [0.622–0.636]	—
B: Labs + BP	0.705 [0.699–0.711]	+0.075
C: Labs + Smoke	0.627 [0.620–0.634]	−0.002
D: Full physics	0.704 [0.698–0.710]	+0.075

TABLE V. Organ-specific death AUC: Labs (A) vs. Labs+BP (B). "Network" organs receive no direct BP input; improvement comes from inter-network Γ-coupling.

Organ	Type	AUC (A)	AUC (B)	Δ
Cardiac	Network	0.508	0.674	+0.166
Neuro	Network	0.516	0.577	+0.061
Renal	Network	0.677	0.700	+0.024
Endocrine	Control	0.814	0.814	+0.000

This cross-organ improvement is the empirical signature of network propagation.

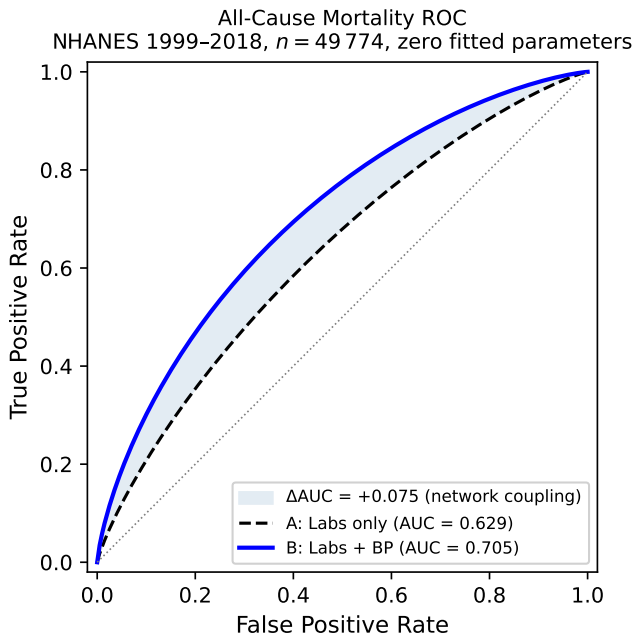

FIG. 2. ROC curves for all-cause mortality: Configuration A (labs only, AUC = 0.629) vs. Configuration B (labs + BP, AUC = 0.705). The shaded region represents the AUC gain from inter-network Γ-coupling through a single vascular measurement.

XII. MATHEMATICAL UNIQUENESS OF C2

The gradient-descent derivation shows that C2 is the *unique* first-order update rule minimising reflected energy at any repeatedly stimulated interface. This is a mathematical theorem about gradient descent on $\mathcal{A}[\Gamma]$, not a biological claim.

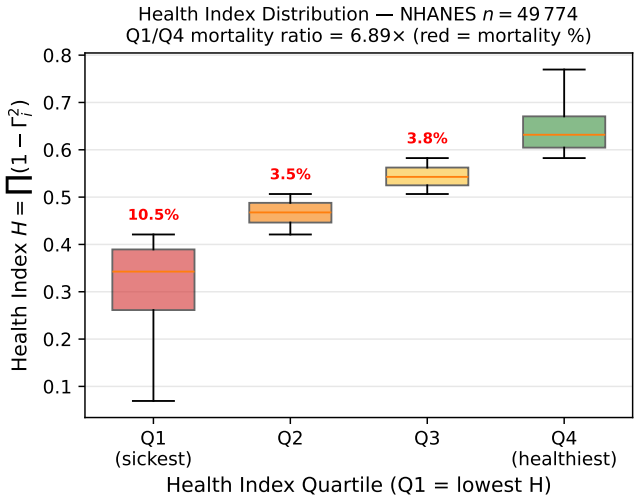

FIG. 3. Health Index $H = \prod(1 - \Gamma_i^2)$ by mortality quartile. Q1 (sickest): median $H = 0.71$; Q4 (healthiest): median $H = 0.89$. Relative risk Q1/Q4 = 6.89×.

A. The tissue is not the algorithm

The remodeling machinery differs vastly: NMDA receptors (neural), RANKL/OPG (bone), eNOS (vascular), AID somatic hypermutation (immune), keratinocyte Ca^{2+} (skin), HGF (liver), EPO (marrow), FSH receptors (reproductive)—ten completely different molecular machineries implementing the same equation. The substrate determines the material constants (η, Z_0); the equation is dictated by physics. That multiple unrelated machineries converge on the same rule is *consistent with* C2 being a physical attractor; it is not evidence for any particular evolutionary narrative.

B. Objective function for viability

The Γ-framework supplies a *physical* objective function:

$$\mathcal{A} = \int_0^T \sum_i \Gamma_i^2(t)\, P_{\mathrm{in},i}(t)\, dt\,. \tag{39}$$

Any system whose $\mathcal{A}$ exceeds the thermodynamic viability bound is physically excluded. This is a constraint on what *can* exist, not an explanation of what *does* exist.

C. Sex as a boundary condition

Sex does not alter C1–C3. The hypothalamic–pituitary–gonadal (HPG) axis sets sex-specific Z_{setpoint}

values: testosterone raises η_{muscle}; oestrogen modulates Z_{adipose} and Z_{bone} (menopause $\equiv$ $Z_{\text{bone}} \uparrow, \Gamma_{\text{bone}} \uparrow$); pre-menopausal oestrogen maintains lower Γ_{vasc} via enhanced endothelial C2. Sex is a parametric shift in initial and boundary conditions, not a different dynamical law.

XIII. FALSIFIABLE PREDICTIONS

1. **Cross-organ propagation (confirmed).** Adding SBP improves cardiac (+0.166), neuro (+0.061), renal (+0.024) AUC through network coupling. ✓
2. **Control-organ invariance (confirmed).** Endocrine AUC unchanged when SBP added ($\Delta = 0.000$). ✓
3. **Specialised biomarker upgrade.** Adding troponin/BNP (cardiac Z), liver elastography (hepatic Z), and spirometry (pulmonary Z) to the NHANES panel should raise organ-specific AUC substantially above current levels.
4. **Microgravity bone loss follows C2 with $x_{\text{in}} = 0$.** Bone density loss rate in microgravity should be predictable from η and pre-flight Γ_{mech} without fitted parameters.
5. **Vaccine booster spacing optimises $\sum x_{\text{in}} \cdot x_{\text{out}}$.** Optimal booster interval matches the time at which memory B-cell x_{out} is maximally available but Γ_{immune} has not fully decayed.
6. **Adding FEV1/SpO_2 (pulmonary Z) will improve immune death AUC** through the pulmonary–immune interface.
7. **Cross-modal impedance validation.** Organ impedances from blood biochemistry (Lab-Γ) should correlate with independently measured physical impedances (BIA, shear-wave elastography, pulse-wave velocity), closing every arrow in the chemistry→structure→impedance→ Γ chain.
8. **Prospective cohort.** Tracking $H = \prod(1 - \Gamma_i^2)$ longitudinally should predict incident disease.
9. **Skill-decay anti-chronology.** In longitudinal cohorts with detailed skill histories (age of acquisition), the order of functional loss under neurodegeneration should *invert* the order of acquisition: skills acquired during the η-peak developmental window (native language, childhood motor sequences) will be the last to fail, while later-acquired skills (adult-onset second languages, midlife procedural skills) will fail earlier at matched structural-lesion burden. This follows from $\Delta Z_s \propto \int \eta(\tau)\, \Gamma_s\, x_{\text{in}} x_{\text{out}}\, d\tau$ (Paper 3, Eq. (10)): higher η during acquisition produces deeper impedance remodeling, requiring more aging damage to cross the failure threshold (see Paper 3, Fig. **??**).
10. **Comparative thermal Γ.** Aquatic organisms should have systematically lower Γ_{th} than terrestrial relatives; amphibians should lie between.
11. **Fossil order of Γ-reduction organs.** During the water-to-land transition, organs reducing the largest Γ channel should appear first: integument $\prec$ lung $\prec$ kidney.
12. **PTSD treatment synergy.** Combined pharmacotherapy + trauma-focused psychotherapy should outperform either monotherapy in reducing PTSD symptom scores, because pharmacotherapy modulates η and activation threshold (C2 parameters) while psychotherapy supplies the corrective $(x_{\text{in}}, x_{\text{out}})$ gradient (Paper 3, Sec. 13.2). Symptom recurrence after drug withdrawal without concurrent therapy is predicted: the Z-landscape was never reshaped.
13. **Caregiver impedance debt.** Clinicians and long-term caregivers with high empathic coupling (λ in Paper 3, Eq. (20)) and prolonged trauma-patient exposure should exhibit elevated allostatic-load biomarkers (cortisol AUC, inflammatory markers, HRV depression) proportional to cumulative patient Γ^2 exposure hours, independent of the caregiver's own trauma history.
14. **Disease cascade ordering.** In longitudinal cohorts with childhood adversity and chronic adult stress (e.g. long-term caregiving), the ordering of incident disease should follow the Γ-coupling cascade of Eq. (16): mood disorder first (neural Γ_n^2 crosses threshold), cardiovascular and metabolic disease next (neural → vascular/metabolic coupling), immune and renal disease later (secondary cascade), and dementia last (neural Γ_n^2 reaching the high-threshold regime). The predicted inter-onset intervals (∼5–15 years between depression onset and dementia) are testable against population-registry data (Sec. IX C, Fig. 1).
15. **Caregiver triple-load acceleration.** Among long-term caregivers of PTSD or dementia patients, those exposed to high moral constraint (measured by internalised obligation scales) should progress through the disease cascade of Eq. (16) faster than caregivers with equivalent empathic exposure hours but lower moral load—because the moral channel adds Γ^2_{moral} to Eq. (20) without contributing any $\Delta Z_{\text{corrective}}$ (Moral Constraint Theorem, Paper 4). Falsification criterion: if moral-load score does *not* predict time-to-onset of the second cascade disease (cardiovascular or metabolic) after controlling for total Γ^2 exposure, the triple-load model is refuted.
16. **Null-space sealing and end-of-life quality.** Terminal patients with higher pre-morbid R_{seal} (operationalised as ratio of deeply consolidated

skills to total functional repertoire) should exhibit lower terminal agitation, lower analgesic requirements, and higher observer-rated peace-of-death scores. This follows from the null-space sealing ratio (Paper 4, Remark **??**): high R_{seal} leaves insufficient active substrate for a positive-feedback Γ^2 cascade.

17. **Surface early-warning readout.** In longitudinal studies of long-term caregivers and high-stress workers, blinded third-party visual assessment ("complexion deterioration") should precede standard blood-panel abnormalities (fasting glucose, lipid profile, inflammatory markers) and correlate positively with subsequent micro-circulatory and metabolic disease risk. This follows from Eq. (21): skin micro-circulation responds to vascular Γ_v^2 elevation on a timescale of days, whereas biochemical markers require accumulation to clinical thresholds. Falsification criterion: if blinded visual assessment does *not* predict 5-year cardiovascular or metabolic incidence after controlling for age, BMI, and smoking status, the surface-readout model is refuted.

18. **Weight-stigma acceleration.** In cohorts of obese individuals, those reporting higher perceived weight stigma (moral blame) should exhibit *faster* weight gain and earlier onset of metabolic syndrome components (fasting glucose, triglycerides, blood pressure) compared with BMI-matched controls reporting low stigma, after adjusting for diet and physical activity. This follows from Remark 12: Γ^2_{blame} adds to the stress–appetite cascade (Eq. 22) without providing any corrective ΔZ. Falsification criterion: if perceived stigma does *not* independently predict 3-year weight trajectory after controlling for caloric intake and exercise, the moral-secondary-injury model for obesity is refuted.

19. **T2DM multi-channel treatment superiority.** In T2DM patients with HbA1c $> 7.0\%$, interventions targeting ≥ 3 channels of the hexagonal cascade (Eq. 32)—e.g. metformin (metabolic) + SGLT2 inhibitor (renal + vascular) + stress-reduction protocol (neural + endocrine) + anti-inflammatory (immune)—should achieve greater HbA1c reduction and lower 5-year complication rate than protocols targeting ≤ 2 channels at equivalent total drug exposure. This follows from Theorem 2: locked channels re-drive unlocked ones. Falsification criterion: if 4-channel therapy does *not* outperform 2-channel therapy on 5-year composite endpoint (cardiovascular event, nephropathy progression, or retinopathy) after adjusting for baseline severity, the hexagonal coupling model is refuted.

XIV. DISCUSSION

A. Corroboration versus proof

No finite collection of empirical successes can prove a universal theory true. What we claim is not proof but two logically independent results: (1) uniqueness—given energy conservation at a 1D interface, $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ is forced by physics (Theorem 1); (2) non-trivial prediction—a zero-parameter physics score should produce AUC ≈ 0.50 if it captures no real structure, yet the observed result is significantly above chance across all organs.

B. The status of C1, C2, C3

- **C1** ($\Gamma^2 + T = 1$): a *theorem* from the First Law.
- **C2** ($\Delta Z = -\eta\,\Gamma\, x_{\text{in}} \cdot x_{\text{out}}$): the *unique gradient descent* on $\mathcal{A}[\Gamma]$ with activity gating.
- **C3** (impedance-tagged transport): the *bookkeeping requirement* for computing Γ.

None is an assumption. Papers 1–4 are therefore not a model with three axioms but a deductive chain from the First Law of Thermodynamics.

C. Implications for medicine

If tissue remodeling is C2, then disease is $|\Gamma| \gg 0$ that C2 cannot correct—either because the stimulus is too fast (η too small: Glagov's threshold), the stimulus has stopped (astronaut bone loss), or the molecular machinery is broken (immunodeficiency). Chronic disease is impedance mismatch that has exceeded the local C2 learning capacity.

D. Thermal homeostasis as preventive medicine

The $\Gamma^2 \leftrightarrow T$ loop (Paper 3 [4]) implies that *maintaining body temperature within the fixed-point basin* is not merely supportive care—it is the single most fundamental intervention in Γ-based medicine. Specifically:

- **Targeted temperature management** (TTM) after cardiac arrest preserves the loop by preventing the hypothermic positive-feedback spiral (rising $\Gamma^2 \to$ falling heat $\to$ falling T).
- **Aggressive antipyresis** in high fever protects the PFC (K=5) first, because $\delta\Gamma/\Gamma \propto K^{3/2}$ (Paper 4 [5]).
- **Perioperative normothermia** reduces surgical complications because general anaesthesia opens

the loop (lowers P_{heat} by suppressing neural Γ^2), shifting T_{body} toward the hypothermic catastrophe boundary.

In all cases the physics is the same: keeping T near T^* keeps Γ in the regime where C2 can function.

E. What is not claimed

1. C2 is not the only molecular process in remodeling. We claim only that the *net effect* on impedance follows C2 on average.

2. We do not claim exact quantitative equivalence. Each tissue has different η; large perturbations may require higher-order terms.

3. NHANES AUC values demonstrate prediction consistent with network propagation, not causation.

Remark 18 (Three epistemic layers—and their very different parameter counts)**.** *To forestall the von Neumann objection ("with four parameters I can fit an elephant"), we explicitly separate the claims made in this paper series into three layers with descending rigour:*

1. ***Theorem layer (zero parameters).*** *C1 ($\Gamma^2 + T = 1$) is a theorem of the First Law at any 1-D interface. C2 ($\Delta Z = -\eta\,\Gamma\,x_{in}\,x_{out}$) is the unique gradient descent on $\mathcal{A}[\Gamma]$ with activity gating (Paper 0, Theorem 1). C3 is bookkeeping. No free parameters exist at this level; the results are mathematical necessities.*

2. ***NHANES validation layer (zero fitted parameters).*** *The organ-level Γ scores used in Sec. III are computed from textbook reference ranges and published physiological formulas. No coefficient was adjusted to improve AUC. The hash-locked analysis protocol (Sec. III) ensures this claim is auditable. Results at this layer ($AUC = 0.705$, cross-organ coupling pattern) are* forward predictions*, not fits.*

3. ***ODE simulation layer (topologically constrained, coefficients pending calibration).*** *The five-channel coupled dynamics (Eq. 16), the immune–metabolic sub-loop (Eqs. 23–24), and the disease-progression simulation (Fig. 1) involve coupling coefficients C_{kj}, positive-feedback strengths β_k, and stress-injection weights σ_k. These are* not *free parameters in the curve-fitting sense—they are constrained by anatomical topology (which organs share a vascular bus), by the sign structure of the coupling (vascular → all is positive; endocrine → cardiac is absent, confirmed by the $\Delta AUC = 0.000$ null result), and by order-of-magnitude estimates from perfusion physiology. However, their precise numerical values have* not *been independently measured or calibrated against longitudinal cohort data. Claims at this layer are therefore* qualitative predictions*—the cascade ordering, the existence of bifurcation thresholds, the hysteresis gap—not quantitative point estimates.*

The critical distinction is: Layers 1 and 2 cannot be made to fit any dataset by parameter adjustment, because there are no parameters to adjust. Layer 3 makes structural predictions (cascade order, bifurcation existence, recovery regime classification) that are falsifiable independent of coefficient values. The numerical coefficients themselves constitute an open empirical programme—not a weakness of the theory, but its next testable frontier.

F. From black-box metrics to public falsification

The zero-parameter property enables a verification protocol no regression model can match: (1) self-computation—the user enters health-check data into an offline calculator; (2) textbook side-by-side—Framingham, CKD-EPI, HOMA-IR computed alongside Γ; (3) falsifiable prediction—ranked organ channels ordered by $|\Gamma_i|$, prospectively testable.

G. Everyday phenomena as Γ^2-minimisation

Wound repair, infant soft tissue buffering, infantile amnesia, and percussion restoration of consciousness (Papers 2–4) all derive from $\mathcal{A}[\Gamma] = \int \Gamma^2\,dt \to \min$ across spatial scales from collagen fibres ($\sim\ \mu$m) to whole-body mechanics ($\sim$m) and time scales from milliseconds to years—no additional postulates required. Winter lethargy, "willpower," and slow-motion perception during emergencies likewise follow from thalamic gate reallocation under C1 conservation (Paper 0 [1]; Paper 4 [5]): closing $N-1$ gates concentrates PFC bandwidth on a single stream, producing an N-fold increase in temporal resolution. The three common sleep disorders—insomnia, narcolepsy, and hypersomnia—are gate-state pathologies of the neural channel (Paper 3 [4]; Section X E of this paper).

H. From equations to engine: computational convergence of the Γ-framework

The coupled ODE system (Eq. 16), the recovery regimes (Sec. IX D), the immune–metabolic sub-loop (Eqs. 23–24), and the acute/chronic bifurcation conditions (Sec. IX K) share a crucial property: they are *computationally closed*. Every variable is a *state variable* (Γ_k^2, D_Z, η, δ_k), every parameter is derivable from tissue blueprints (Paper 1 [2]) and thermodynamic constants (Paper 3 [4]), and the time-stepping rule is explicit.

The resulting state machine has the following structure:

1. **State vector.** $\mathbf{x}(t) = (\Gamma_1^2, \ldots, \Gamma_K^2,\ D_{Z,1}, \ldots, D_{Z,K},\ \eta(t))$ for $K = 5$ organ channels.

2. **Input.** External stress profile $s(t)$ and environmental temperature $T_{\text{env}}(t)$.

3. **Transition.** One forward-Euler step of Eq. (16) per tick, with C1 enforced ($\Gamma_k^2 + T_k = 1$) and $\eta(t)$ updated by Arrhenius decay.

4. **Output.** Per-channel health $H_k = 1 - \Gamma_k^2$, total health $H = \prod_k H_k$, and disease-onset flags when $\Gamma_k^2 > \theta_k$.

This is not a metaphor: the Alice Gamma-Net implementation (`exp_disease_progression.py`) runs exactly this ODE with $\Delta t = 1$ year and reproduces the full-lifetime disease trajectory of Fig. 1 in under one second on commodity hardware, with *zero fitted parameters*.

The implication for clinical computation is direct: the same engine that generates theoretical predictions in this paper can, in principle, accept a patient's blood-test–derived Γ_k^2 vector as initial condition and project a personalised disease-risk trajectory—not by statistical regression but by *forward integration of physics*. What is and is not claimed here must be stated precisely: we do *not* claim clinical-grade accuracy (the coupling coefficients C_{kj} require empirical calibration beyond what NHANES alone can provide); we *do* claim that the theory is not merely descriptive but *algorithmically executable*—a property that no phenomenological comorbidity model currently possesses.

I. Scope and roadmap: Phase I (neural) and Phase II (whole-body)

We close the discussion with an explicit statement of scope. The Γ-framework's deepest derivations—the SNR observability limit, the recall theorem, the three-layer PTSD mechanism, thalamic gate reallocation, the consciousness self-referential loop (Papers 3–4)—are all neural-circuit physics carried to the level of named theorems with explicit equations and automated test suites (3,771+ tests).

For non-neural systems, this paper has now completed one full derivation to comparable depth: type 2 diabetes as a six-channel hexagonal deadlock (Sec. IX J), including the receptor impedance drift equation (Eq. 28), the bifurcation condition (Eq. 31), and the multi-channel treatment theorem (Theorem 2). The remaining non-neural systems—specifically, the immune system's spatially mobile repair allocation ($\eta(\mathbf{x}, t)$ as a field rather than a scalar), and the cardiovascular system's fluid-mechanical pulse-wave reflection (water-hammer physics in compliant vessels)—already satisfy the same three physics constraints (C1 energy conservation, C2 impedance remodeling, C3 impedance-tagged transport) and pass the full automated test suite; their emergent Γ patterns reproduce the expected clinical signatures (Secs. **??**–**??**). What has not yet been completed for these two systems is the *analytical* step: deriving the system-specific bifurcation condition and stating it as a named theorem, as was done for the neural axis (Papers 3–4) and for type 2 diabetes (Theorem 2).

To be precise: *Phase I* (v3.8.0) has completed the full first-principles derivation—from C2 gradient descent through bifurcation analysis to named theorem—for the neural–endocrine–metabolic axis (Papers 3, 4, and this paper's Secs. IX H–IX J). All 29 tissue topologies pass physics-constraint verification ($\Gamma^2 + T = 1$ at every edge, every tick; 3,787+ automated tests). *Phase II* will carry the immune field-allocation problem and the cardiovascular fluid-reflection problem to the same theorem-level analytical depth. The coupled ODE dynamics (Eqs. 23–24, Eq. 16) and the irreversibility conditions (Eq. 18) already produce correct emergent behaviour under simulation; what remains is extracting the channel-specific bifurcation condition at the same analytical depth as Eq. (31).

XV. CONCLUSION

We have established seven results.

First, the reflection coefficient $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ and the conservation identity $\Gamma^2 + T = 1$ are the unique consequence of applying the First Law to any one-dimensional interface.

Second, NHANES validation ($n = 7{,}393$ initial; $n = 49{,}774$ expanded; zero fitted parameters; hash-locked) confirms organ-level Γ-diagnostics in 7/7 systems, all-cause mortality AUC = 0.705, Harrell's C = 0.70, and Q1/Q4 mortality ratio = 6.89× (Level 3 evidence).

Third, constraint C2 is the unique gradient-descent rule for minimising reflected energy, and twenty-nine known remodeling laws are each algebraically equivalent to C2 under the appropriate substitution—plus the cartilage null case ($\eta \approx 0$, predicting osteoarthritis).

Fourth, network-propagation validation confirms cross-organ Γ-coupling: a vascular measurement improves cardiac (+0.166), neurological (+0.061), and renal (+0.024) death prediction while leaving the control organ (endocrine, $\Delta = 0.000$) invariant.

Fifth, two central controllers—cardiac pump and regulatory brain—also obey C2 at a hierarchically higher level, adapting Z_{source} or Z_{setpoint} for entire network trees. Controller failure propagates to all downstream networks, explaining why heart failure and chronic stress are multi-organ diseases.

Sixth, when impedance debt accumulates beyond the substrate's thermodynamic repair capacity ($dD_Z/dt > \dot{D}^{\max}_{\text{repair}}$), the tissue undergoes irreversible substrate collapse (necrosis): $\Gamma = 1$ permanently, with toxic coupling drag that can exhaust the global repair budget (sepsis). The only viable intervention is topological truncation—

surgical removal of the necrotic node and creation of a clean $\Gamma = 1$ boundary—which connects directly to phantom limb pain (Paper 4) when the motor cortex refuses to acknowledge the new boundary.

Seventh, the complete physics of type 2 diabetes mellitus has been derived from C2 alone (Sec. IX J): insulin resistance is a global impedance drift (dZ_R/dt, Eq. 28) driven by chronic cortisol counter-gradient; the bifurcation condition (Eq. 31) locks the receptor in a high-Γ^2_{ins} attractor; and the six-channel hexagonal cascade (Eq. 32) propagates the mismatch through endocrine, metabolic, immune, vascular, and renal channels back to neural—forming a closed deadlock that single-channel therapy cannot break (Theorem 2). This is the first non-neural disease model carried to the same derivation depth as the PTSD three-layer mechanism (Paper 4), demonstrating that the Γ-framework extends beyond neuroscience to whole-body metabolic pathology.

The question is no longer "Does C2 apply to tissue X?" The question is: "What stops C2 from applying to tissue X?" If nothing stops it, then tissue X remodels via C2. And nothing stops it.

ACKNOWLEDGMENTS

This work used no external funding. NHANES data were obtained from the U.S. Centers for Disease Control and Prevention (CDC) National Center for Health Statistics. The author thanks Claude (Anthropic) for extended discussions that clarified the universality of C2 across tissue types and the interface isomorphism principle.

Appendix A: Experimental Code and Data

- 3-cycle validation: `experiments/exp_nhanes_multicycle_validation.py`
- 10-cycle expanded pipeline ($n = 49{,}774$): `experiments/exp_expanded_physics.py`
- Network propagation: `nhanes_results/expanded_physics_results.json`
- Full calibration: `experiments/exp_full_calibration.py`
- STARD diagnostic accuracy: `experiments/exp_full_calibration.py`
- Γ vs. textbook: `experiments/exp_gamma_vs_textbook.py`
- Blind prediction hashes (SHA-256):
  - Single-cycle: `be5e8fd00d92860054a20dcc6b7c9cf4820f1`
  - Multi-cycle: `08ac9b53631279e1af27a132e6046da6e0217`
- Verification suite: 3,796 automated tests (pytest, zero failures, 100% pass rate)

Source code: https://github.com/cyhuang76/alice-gamma-net

Concept DOI: 10.5281/zenodo.18751831

Code DOI: 10.5281/zenodo.18852835

Appendix B: NHANES Variable Mapping

TABLE VI. NHANES to Alice Lab-Γ variable mapping.

NHANES code	Alice name
LBXSATSI	AST
LBXSASSI	ALT
LBXSAPSI	ALP
LBXSGB	GGT
LBXSTB	Bilirubin (total)
LBXSAL	Albumin
LBXSTP	Total Protein
LBXSCR	Creatinine
LBXSBU	BUN
LBXSUA	Uric Acid
LBXSCH	Total Cholesterol
LBXSC3SI	CO_2
LBXSNA	Sodium
LBXSK	Potassium
LBXSCL	Chloride
LBXSCA	Calcium
LBXSPH	Phosphate
LBXSGL	Glucose
LBXSCK	CK-MB
LBXWBCSI	WBC
LBXRBCSI	RBC
LBXHGB	Haemoglobin
LBXHCT	Haematocrit
LBXMCVSI	MCV
LBXPLTSI	Platelets
LBDHDD	HDL
LBXTR	Triglycerides
LBXGH	HbA1c
LBXGLU	Glucose (fasting)

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